Flexible Textile-Based Antenna for Next Generation Wearable Health Monitoring

Submitted in partial fulfillment of the requirements for the degree of

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and
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November, 2025

DECLARATION

I hereby declare that the thesis entitled "Flexible Textile-Based Antenna for Next Generation Wearable Health Monitoring" submitted by me, for the completion of the course "BECE305L – Project" to the School of Electronics Engineering, Vellore Institute of Technology, Vellore, is bonafide work carried out by me under the supervision of Dr. Rajkishor Kumar.

I further declare that the work reported in this thesis has not been submitted previously to this institute or anywhere.

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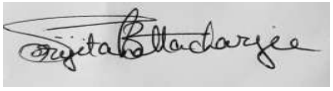

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EXECUTIVE SUMMARY

Through this project, we perform the designing and performance evaluation of a flexible microstrip patch antenna that operates in the 2.5 GHz ISM Band, aimed at wearable biomedical and wireless body area network applications. We have simulated the antenna in two different configurations - free space as well as on a human body phantom using CST Studio Suite. These configurations help us analyse the effects of body loading on the impedance, gain, efficiency, and safety associated with the antenna.

In free space, the antenna achieved a resonant frequency of 2.998 GHz, a return loss of -12.77 dB, bandwidth of 100MHz, and a peak gain of 3.03 dBi. When placed on the body model, the antenna shows a downward frequency shift to 2.97 GHz, along with a degraded return loss of -11.09 dB, a reduced gain as well as a lower efficiency due to the dielectric loading and power absorption seen by the body tissues.

Power loss density analysis at 2.5 GHz highlights the energy absorption near the skin-patch interface but it was seen to remain within safe operational limits. The antenna retained stable broadside radiation and acceptable impedance matching which validates its suitability for wearable use.

Future improvements such as AMC/EBG backing, air-gap spacing, or feed re-optimization can be considered to help minimize losses and enhance safety. The study confirms the antenna's promise for next-generation health-monitoring wearables.

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ABBREVIATIONS

Abbreviation	Full Form
WABN	Wireless Body Area Network
AMC	Artificial Magnetic Conductor
EBG	Electromagnetic Bandgap
FBR	Front-to-Back Ratio
ISM	Industrial, Scientific, and Medical Band
SAR	Specific Absorption Rate
SIW	Substrate Integrated Waveguide
SMA	SubMiniature version A Connector
S_{11}	Return Loss Parameter
VSWR	Voltage Standing Wave Ratio

CST	Computer Simulation Technology

SYMBOLS AND NOTATIONS

Symbol/Notation	Parameter
εr	Relative Permittivity (Dielectric Constant) - “r” denotes relative property.
$\tan \delta$	Loss Tangent - represents the dielectric loss tangent of the material.
σ (S/M)	Electrical Conductivity
Z_0	Characteristic Impedance
S_{11}	Return Loss or Reflection Coefficient at Port 1.
η	Efficiency
ηrad	Radiation Efficiency
ηtot	Total Efficiency
$E\theta$, $E\varphi$	Electric Field Components in the spherical coordinates.
$H\theta$, $H\varphi$	Magnetic Field Components in the spherical coordinates.
SARavg	Average Specific Absorption Rate within tissue volume.

1. INTRODUCTION

1.1 LITERATURE REVIEW :

In today's scientific scenario, Wireless Body Area Networks or WBAN are gaining a lot of traction. These can be defined as being a network of sensor nodes which are attached to the body in order to measure and monitor bio signals (such as heart rate, blood pressure, brain waves, etc). In this landscape, wearable antennas are becoming popular components of WBAN because of their portable and easy-to-use nature. These antennas, however, need very stable electromagnetic performance in order to achieve maximum positive results while transmitting bio-signals. The human body is made of components which have water and electrolytes present in them. As such, it is a lossy dielectric medium which has a negative effect on the antenna's performance which can be seen in phenomena such as detuning (when the resonant frequency of the antenna gets altered), or when the gain of the antenna decreases due to some of the radiated power being absorbed by the body's tissues. Another major problem is the SAR or the Specific Absorption Rate. It refers to the rate of absorption of electromagnetic energy which gets converted to heat inside the human body. This SAR value should be within a specific range since a high SAR can be unsafe for human exposure.

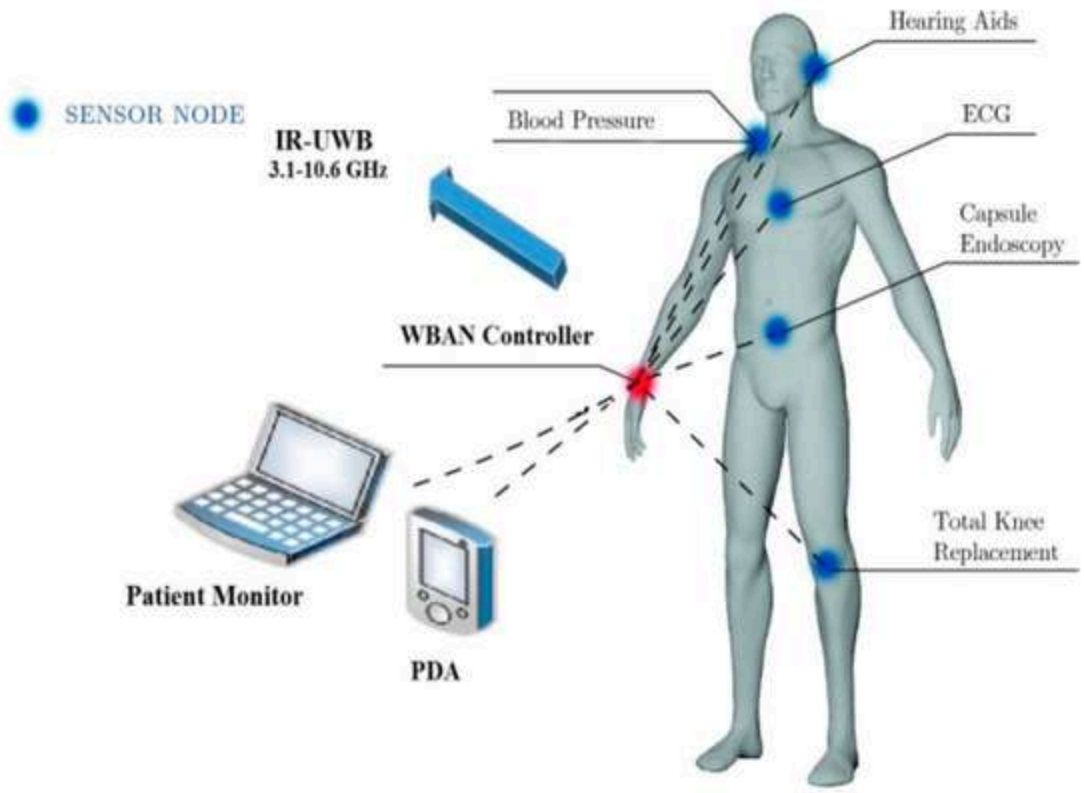


Figure 1 : Wireless Body Area Network.

In the research done by Ashyap et al. (2025), we can see an extensive deep-dive into the methods to improve the efficiency of wearable antennas, while making them more safe and comfortable to wear. They elucidated how incorporating periodic structures such as AMCs (Artificial Magnetic Conductors) and EBG (Electromagnetic Bandgap) surfaces into wearable antennas can lead to their increased efficiency. These structures are basically special layers that are present between the antenna and the human body. They thus act as a shield and prevent unwanted and wasteful energy absorption of energy which is radiated by the antennas. This not only improves the overall efficiency of the antenna but also makes it safe for human use. They also showed how by incorporating these aforementioned structures, wearable antennas can achieve high FBR (front-to-back ratio) and low SAR.

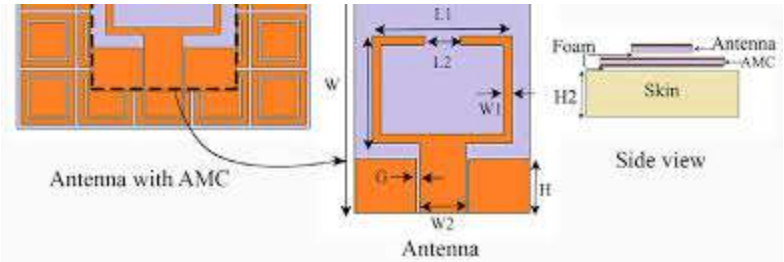


Figure 2 : Wearable Antennas employing the use of AMC layers.

In the same research, the authors also compared different types of textiles to see which one would suit the purpose of the wearable antennas while being comfortable to wear. They found that felt, denim and PDMS (polydimethylsiloxane) were most appropriate for this purpose. They provided low dielectric loss, high electromagnetic stability, and even maintained their shape when bent.

Kao and Chuang (2020) did extensive research into the effects of bending on wearable antennas, particularly those which were fabric-based. They analysed two types of bending - longitudinal (bending of the antenna along the direction of radiation) and transverse (bending of the antenna across the direction of radiation). It was found that bending negatively affects the performance of these antennas since it changes their geometry. However, longitudinal bending caused greater degradation to the antenna's performance than transverse bending. It resulted in higher detuning and lesser gain. The authors showed how the problems associated with bending can be lessened by choosing those materials which have low anisotropy (having the same properties in all the directions) and by optimising the substrate thickness in such a way that bending does not lead to high degradation.

Sandeep et al. (2024) found even more eye-opening insight into the effects of bending on antennas. They designed a circularly polarized textile antenna and then tested its performance at different bending radii. They found that as the bending angles increased, the circular polarization stability got difficult to maintain. However, the maintenance of radiation efficiency can be done by optimising crucial elements of antenna design such as the ground structure and feed positioning.

Rafiq (2024) also reviewed and compared the manufacturing techniques for textile based wearable antennas on the basis of several aspects. The fabrication methods that were taken for consideration were embroidery, screen printing, and inkjet deposition. The findings were quite profound - it was seen how the designs which were embroidery-based resulted in higher durability as well as washability. Further, the printed designs were found to give more precision but did not have high longevity under mechanical stress.

Selecting an appropriate material in order to design the wearable antenna is of the utmost importance since the electrical as well as mechanical properties of the material determines the performance of the antenna. Marterer et al (2024) compared the performance of different textile materials used in designing wearable antennas. The main materials studied were polyester, cotton, and conductive nylon. It was found that the permittivity, moisture absorption capability, and the fabric thickness associated with each fabric were crucial factors influencing their respective performances. They found that materials with lower permittivity, moderate low moisture absorbing capability, and moderate thickness were most suitable for making wearable antennas.

One of the key concerns of wearable antenna design is to improve electromagnetic isolation from the human body. This not only improves the efficiency of the antenna but also ensures the safety of the antenna for wearable purposes. Thangavelu et al. (2024) put forward a wearable antenna design which was aimed at achieving this electromagnetic isolation with the help of two main structures - Artificial Magnetic Conductors (AMCs) and Substrate-Integrated Waveguide (SIW). Simulations were done on cotton, denim, and jute textile substrates and the results were striking. It was found that using the AMCs led to a 30-40% reduction in SAR as compared to other standard antennas. However, integration of the AMC layer did not compromise the gain and bandwidth of the antenna.

Abdelghany et al. (2024) designed a dual-band textile antenna and tested its performance in free space as well as on a multi-layer body phantom (a model that was designed to accurately mimic the tissue layers present in an actual human body). They found that the antenna that they designed maintained a consistent performance (similar radiation patterns, stable impedance matching, and less shifting of frequency) even when tested on the human body phantom. The authors further emphasized on the importance of testing wearable antennas on realistic body models in order to ascertain whether or not they are ready to be used for wearable or biomedical use since the performance of the antennas can change drastically once they are placed on the skin.

The comprehensive literature done in this project highlights the importance of choosing an appropriate material while designing a wearable antenna, incorporating practices that ensure good electromagnetic isolation of the antenna from the human body, enhancing the antenna performance while ensuring user safety, and minimising degradation caused due to bending and mechanical stresses. It was also seen how there is a need for comparative studies between comparative studies between the performance of antennas in free-space conditions and in body-mounted conditions. Efforts have been made to directly address this need in this project.

1.2 RESEARCH GAPS :

1.2.1 Lack of dual-environment validation

We have seen how most of the studies performed in the domain of wearable antenna design, such as those done by Ashyap et al. (2025) and Sandeep et al. (2024) analyse and evaluate the performance of antennas in free space. However, that itself is not enough to determine whether or not the antenna is ready for wearable purposes since mounting the antenna on the human body can cause significant changes to the antenna's performance. Abdelghany et al. (2024) did perform a comparative study between the performance of a wearable antenna in free-space vs a body-mounted environment. However, these studies are few and far between. A direct comparative analysis is the need of the hour since it can be greatly helpful in quantifying what happens to the level of detuning, gain, and SAR once the antenna is placed on the human body.

1.2.2 Limited integration of flexibility and safety optimization

Studies evaluating the performance of wearable antennas often focus only on one particular factor affecting its performance. For eg, Kao and Chuang (2020) and Sandeep et al. (2024) focussed on the effects of mechanical bending on antenna performance and ways to mitigate it while Thungavelu et al. (2024) explored ways to reduce SAR while enhancing the electromagnetic isolation. We can thus see a research gap in jointly optimising all of these aspects - antenna flexibility, efficiency, and SAR compliance, all within a single framework.

1.2.3 Insufficient study of material-dependent performance variation

The study performed by Rafiq (2024) compared how different fabrication methods affect overall antenna performance. While Marterer et al. (2024) focussed on how the different fabric materials impact antenna performance in different ways. However, these aspects were examined individually and not in an integrated or combined way. It can be said that these studies showed that the properties of the material matter as far as antenna performance is concerned, but they did not establish a proper correlation between these properties and the overall antenna performance.

1.2.4 Limited validation of antenna behavior under realistic bending

In most of the existing research on wearable antennas, the effects of bending are only tested for certain specific angles (such as 30° or 60°). However, when a real human body moves, an antenna may bend at numerous angles. As such, studying how bending at only a few specific angles affects antenna performance does not tell us how exactly the antenna will perform on the actual human body.

Moreover, a lot of experiments test antennas on single-layer tissue models which do not accurately represent how antennas will behave when mounted on a human body. For that purpose, multi-layer phantoms must be used.

1.2.5 Need for standardized performance evaluation

There is no particular standard method/framework to comprehensively analyze antenna performance which compares results across different designs. While some researchers might focus only on electromagnetic parameters (such as gain, SAR, etc), others might only focus on mechanical parameters (such as flexibility, comfort, etc). However, this makes it very difficult to ascertain which design is better overall. For example, an antenna with a high gain might have poor flexibility. As such, there is a need to study the electromagnetic and mechanical parameters side by side in a unified way.

1.3 PROBLEM STATEMENT :

Wearable antennas are key components of biomedical devices operating in the realm of WBAN (Wireless Body Area Network). These are required for the efficient transmitting of bio-signals for high quality healthcare monitoring. However, when flexible, textile-based antennas are mounted on the human body, they often face a lot of degradation in their overall performance due to various causes such as bending, stretching, absorption of radiation by the human body, etc. This is a major issue since it can create problems such as detuning, reduction of gain, and an increase in SAR.

We have seen how the already existing research on wearable antennas focus mainly on determining an appropriate material choice or achieving electromagnetic safety and isolation. However, there are limited comparative studies between how the antenna performs in free space versus on the human body which are essential to ascertain if the antenna is ready to be used for wearable purposes.

There are also very few studies which delve into how wearable antennas behave under the kind of bending levels which they would be faced with if placed on the human body for practical purposes.

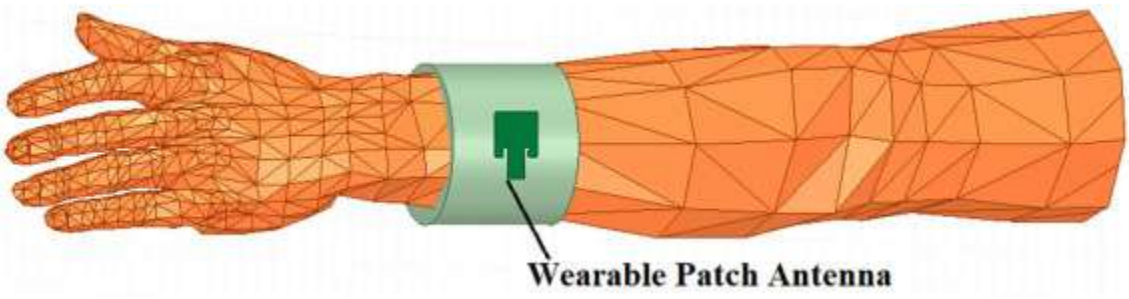


Figure 3 : Wearable Patch Antennas.

This project is aimed at addressing these glaring gaps by designing and simulating a flexible textile antenna on CST Studio Suite software and by further doing a comparative analysis of the antenna's performance in free space and on a body phantom. Different parameters of antenna performance such as the antenna's S11, gain, radiation pattern, SAR, etc will finally be assessed at multiple bending conditions for comprehensive result finding.

1.3.1 RELEVANCE OF THE PROBLEM STATEMENT WITH RESPECT TO SUSTAINABLE DEVELOPMENT GOALS (SDGs)

- SDG 3 - Good Health and Well Being
  In our project, our proposed antenna would promote good health by allowing the continuous monitoring of biosignals. This would, in turn, lead to better and earlier disease detection and lead to enhancement in the overall quality of healthcare.

- SDG 9 - Industry, Innovation, and Infrastructure
  We have incorporated innovative practices in our project by combining flexible materials with smart electronics in order to design our proposed antenna.

- SDG 12 - Responsible Consumption and Production
  Throughout our project, we have tried to emphasize the importance of using sustainable materials. This reduces electronic waste and promotes a green future.

2. PROJECT OBJECTIVE

This project helps us design, simulate and analyze a flexible microstrip patch antenna operating at 2.5 GHz for wearable biomedical and wireless body area network (WBAN) applications. The comparative study focuses on understanding how human body proximity influences key antenna parameters, including resonant frequency, return loss, bandwidth, gain, radiation efficiency and power absorption (SAR).

Using CST Studio Suite, the antenna is modeled and evaluated under two configurations - one is a free space and the other is a multilayer human body phantom consisting of skin, fat, muscle and bone tissues. This helps us observe detuning and performance degradation caused by the dielectric loading caused by the body phantom. The project

also assesses power loss density at 2.5 GHz to quantify the electromagnetic energy absorption and verify safe operating levels for wearable use.

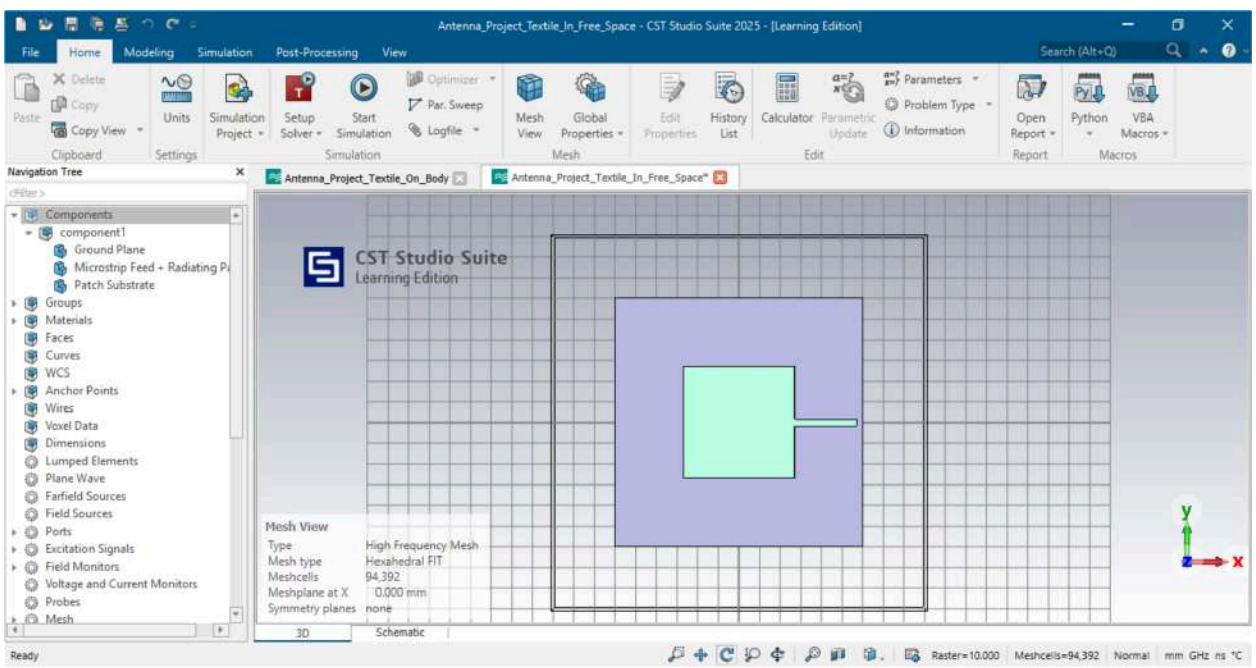
Furthermore, the project helps us explore the possible design optimizations such as Artificial Magnetic Conductor (AMC) backing, air-gap introduction and feed reconfiguration to enhance radiation performance, minimize losses, and ensure the safety of the user. Through this study, a reliable and efficient wearable antenna model is developed, contributing to the advancement of health-monitoring and biomedical telemetry systems.

3. PROPOSED WORK

3.1 DESIGN APPROACH :

The design approach for this project centers on developing a flexible microstrip patch antenna optimized for wearable biomedical applications operating at 2.45 GHz.

First we started with selecting appropriate substrate and conductor materials to ensure both mechanical flexibility and electromagnetic stability. We used the standard transmission-line equations taking into consideration the effective dielectric constant and the fringing effects in order to calculate the antenna dimensions.



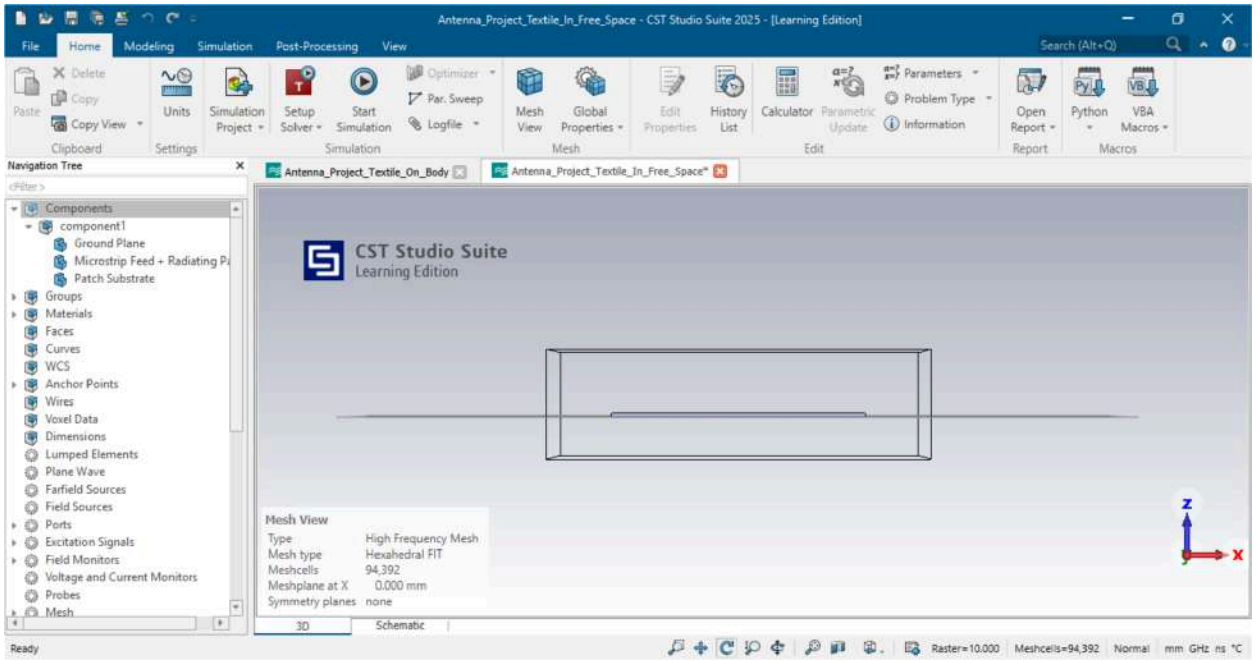


Figure 4 : Rectangular Patch Antenna Simulated in Free Space on CST Studio Suite.

We chose a rectangular patch geometry for its simplicity, ease of fabrication, and stable radiation characteristics. The antenna employs a microstrip feed line to achieve impedance matching while maintaining a compact profile. This model was built and simulated in CST Studio Suite, where different layers including - the radiating patch, the substrate, the ground plane, and a four-layer body phantom were sequentially added.

To evaluate real-world performance, simulations were conducted under two configurations, one in free space and one on the human body phantom. After that, the key parameters required to study the antenna's radiating pattern were analyzed.

3.2 TECHNICAL DESCRIPTION :

The antenna is composed of three primary layers - the radiating patch, the dielectric substrate and the ground plane. For the on-body analysis, we have considered a multilayer human body phantom consisting of four layers - the skin, fat layer, muscle and the bone.

The *radiating patch* is made out of copper, which was chosen for its high electrical conductivity and stability under bending.

The *substrate* is made using a flexible dielectric material with a relative permittivity of 2.2 and a loss tangent of 0.0009, selected to provide low surface-wave loss and high

efficiency.

The *ground plane* is modeled using copper again to ensure efficient reflection and minimize backward radiation.

The *body phantom* comprises four tissue-equivalent layers, each assigned with material properties based on the literature values at 2.45 GHz.

A *microstrip feed line* is used for excitation, allowing impedance matching between the source and the patch. The excitation is applied via a waveguide port placed at the end of the feed line.

An AMC Backing layer comprising 9 small AMC patches, an AMC Substrate and an AMC ground have also been included in the simulation in order to help improve the antenna's overall performance.

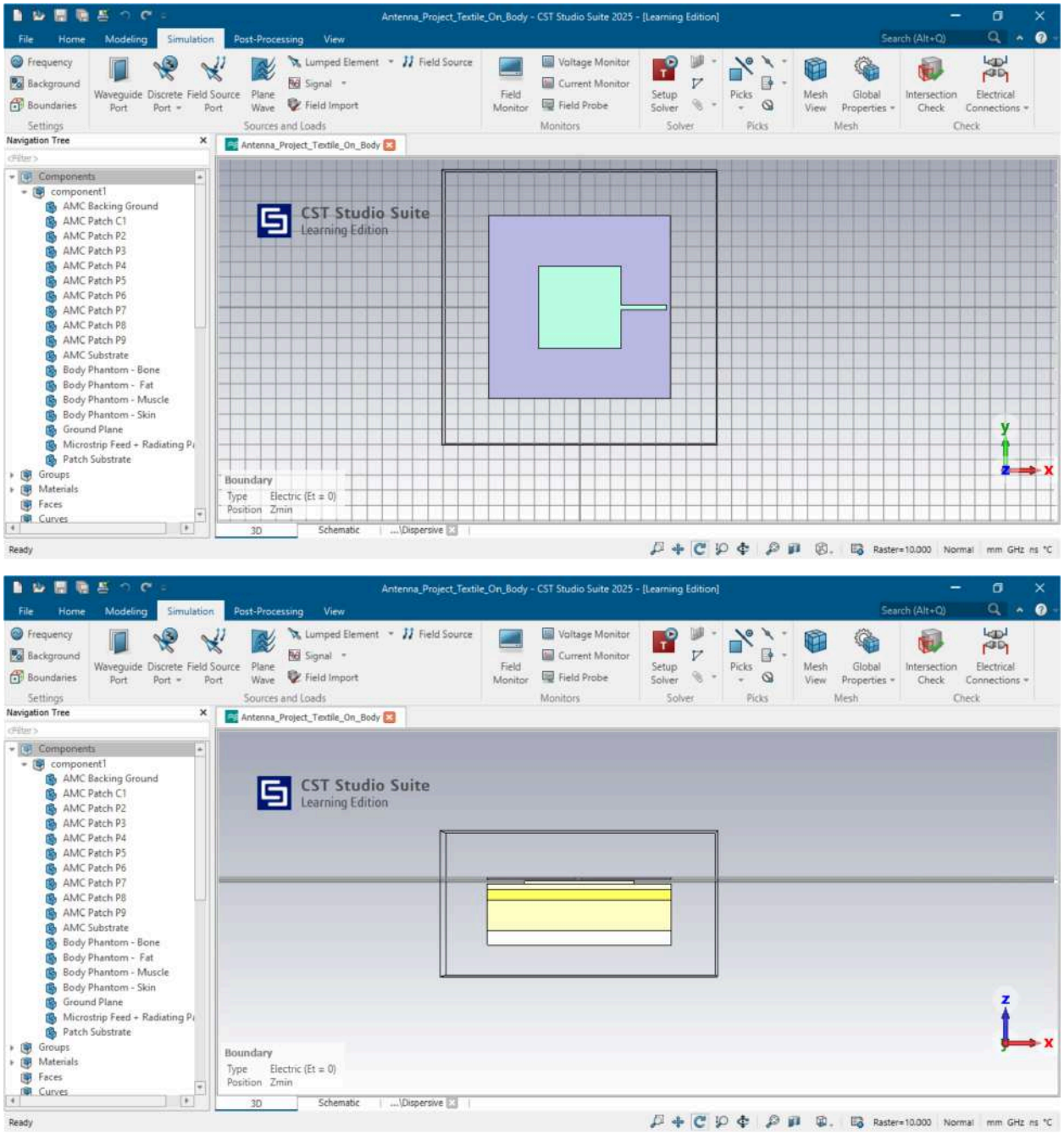


Figure 5 : Rectangular Patch Antenna Simulated on the Body Phantom on CST Studio Suite.

Layer	Material	Thickness
Microstrip Feed + Radiating Patch	Copper (annealed)	0.035 mm
Patch Substrate	Polyimide (lossy)	1.465 mm

Ground Plane	Copper (annealed)	0.035 mm
AMC Patches (9)	Copper (annealed)	0.035 mm each
AMC Substrate	FR-4 (lossy)	1.965 mm
AMC Backing Ground	Copper (annealed)	0.035 mm
Body Phantom - Skin	Custom Material : 1.49 S/m; εr = 37.95	3 mm
Body Phantom - Fat	Custom Material : 0.11 S/m; εr = 5.27	7 mm
Body Phantom - Muscle	Custom Material : 1.77 S/m; εr = 52.67	20 mm
Body Phantom - Bone	Custom Material : 0.82 S/m; εr = 18.49	10 mm

Table 1 : Material Properties and Dimensions used to simulate the Rectangular Patch Antenna.

Then the antenna is simulated using CST Studio Suite in the Time Domain Solver mode, where the boundary conditions are defined as Air (Open). The boundary condition below the ground is set to be defined as Electric.

Performance parameters such as S_{11}, VSWR, Radiation Efficiency, far-field gain, E-Field distribution, and Power Loss Density (SAR) are analysed under both free-space and on-body conditions.

This design emphasises on the flexibility, biocompatibility, and safety, making it idea for integration into wearable health-monitoring systems.

4. HARDWARE/SOFTWARE TOOLS USED

To conduct this comparative analysis, we have considered and performed software simulations of the patch antenna under the two environments using the **CST Studio Suite Software.** The tools available in the software enabled accurate modeling, electromagnetic field computation, and visualization of antenna performance under both the environment conditions.

5. RESULT ANALYSIS

After simulating the constructed patch antenna under both the testing environments, we have taken into consideration the following graphs and plots to help perform our comparative analysis.

1. *S Parameters Graphs (both Cartesian and Real/Imaginary Part Plot) :*
   These plots show the variation of return loss (S_{11}) with frequency. It is essential for identifying the resonant frequency and determining how well the antenna is impedance matched to 50Ω.

   In our simulation, in free space, the antenna resonates at 2.998 GHz with $S_{11} = -12.77$ dB. On body, resonance shifts slightly down to 2.97 GHz with $S_{11} = -11.09$ dB. Both the cases indicate a return loss below -10dB indicates that more than 90% of the power is radiated confirming an efficient energy transfer. The small frequency shift is seen when observing the resonant frequency which confirms the detuning caused by the human body's dielectric loading, which slightly affects impedance matching but remains within acceptable limits for wearable use.

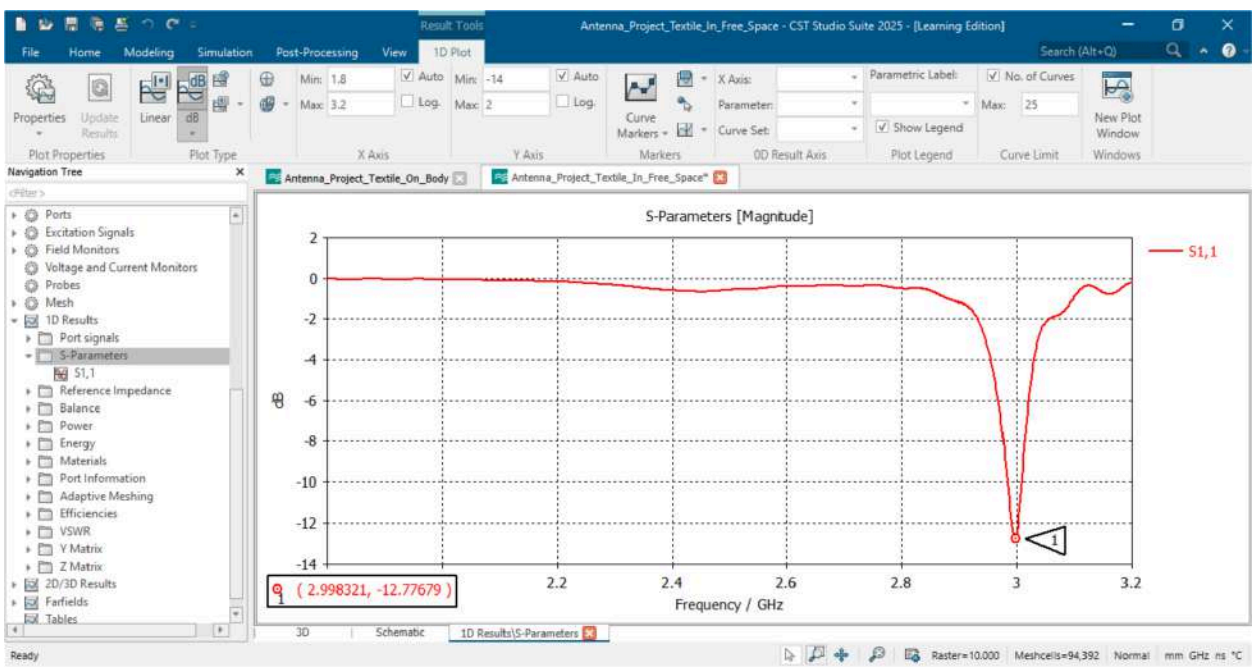


Figure 6 : Cartesian Plot of the S Parameters for the Patch Antenna in Free-Space.

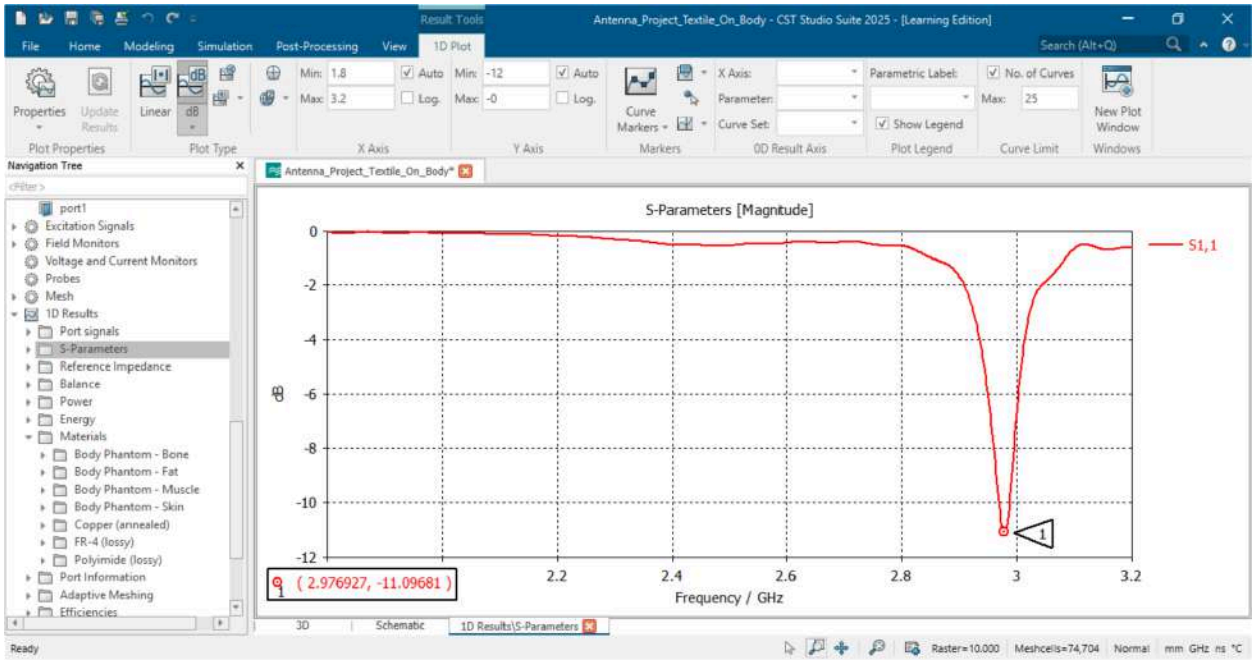


Figure 7 : Cartesian Plot of the S Parameters for the Patch Antenna placed on the Body Phantom.

The real and imaginary components of the S_{11} provide insights into the reactive behavior of the antenna. They help analyse whether the antenna is inductive or capacitive at particular frequencies and assist in fine-tuning the feed or patch dimensions to enhance the matching. At resonant frequency the real part is close to 50Ω while the imaginary part crosses zero, confirming resonance in the ideal situation. When placed on-body, the curve shows a small downward shift, matching the detuned resonant frequency.

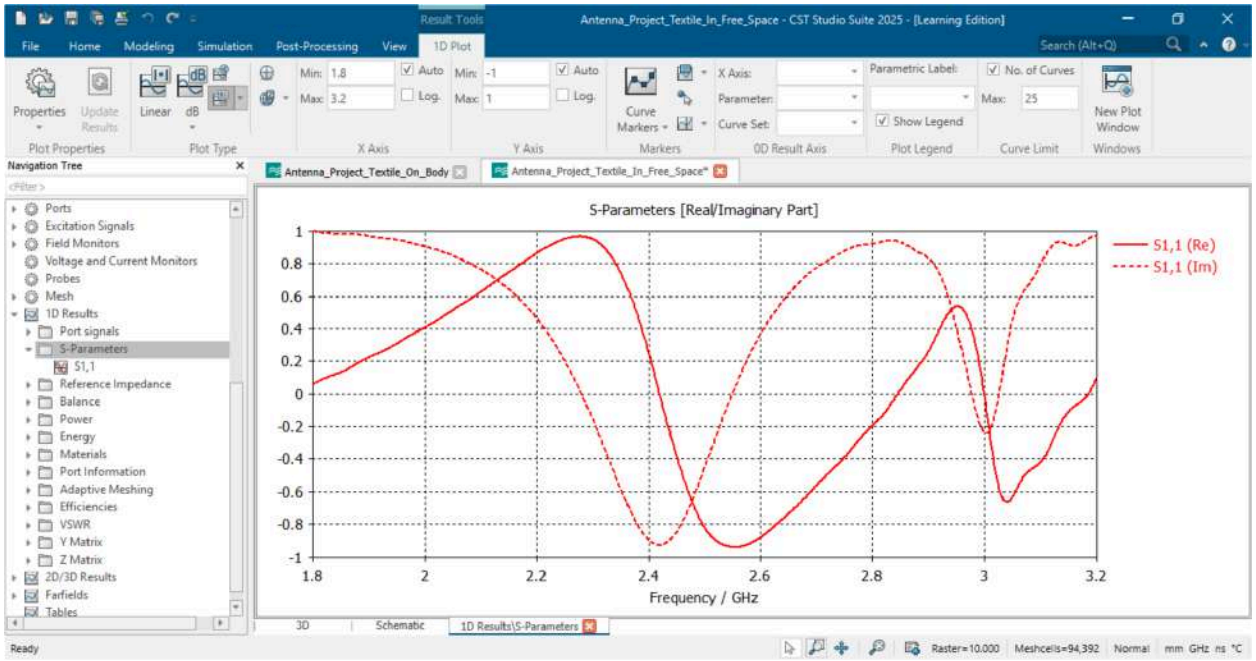


Figure 8 : Real/Imaginary Part Plot of the S Parameters for the Patch Antenna in Free-Space.

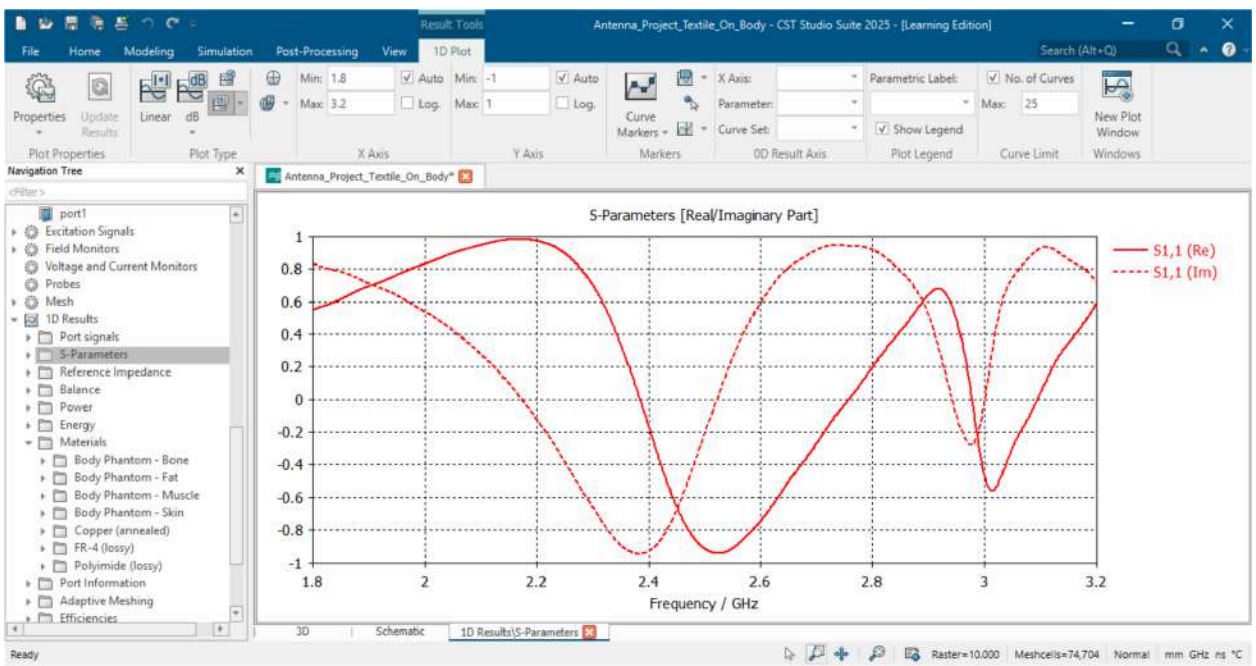


Figure 9 : Real/Imaginary Part Plot of the S Parameters for the Patch Antenna placed on the Body Phantom.

2. *S Parameters Graphs (Smith Chart) :*
   The Smith Chart represents the complex impedance of the antenna as a function of frequency. It helps us visually confirm how close the input impedance is to the center (50Ω normalised impedance). The proximity of the curve to the center denotes good matching and low reflection. In our study, in case of the on-body simulation, the curve moves slightly away from the center, which indicates minor mismatch due to body coupling and dielectric influence.

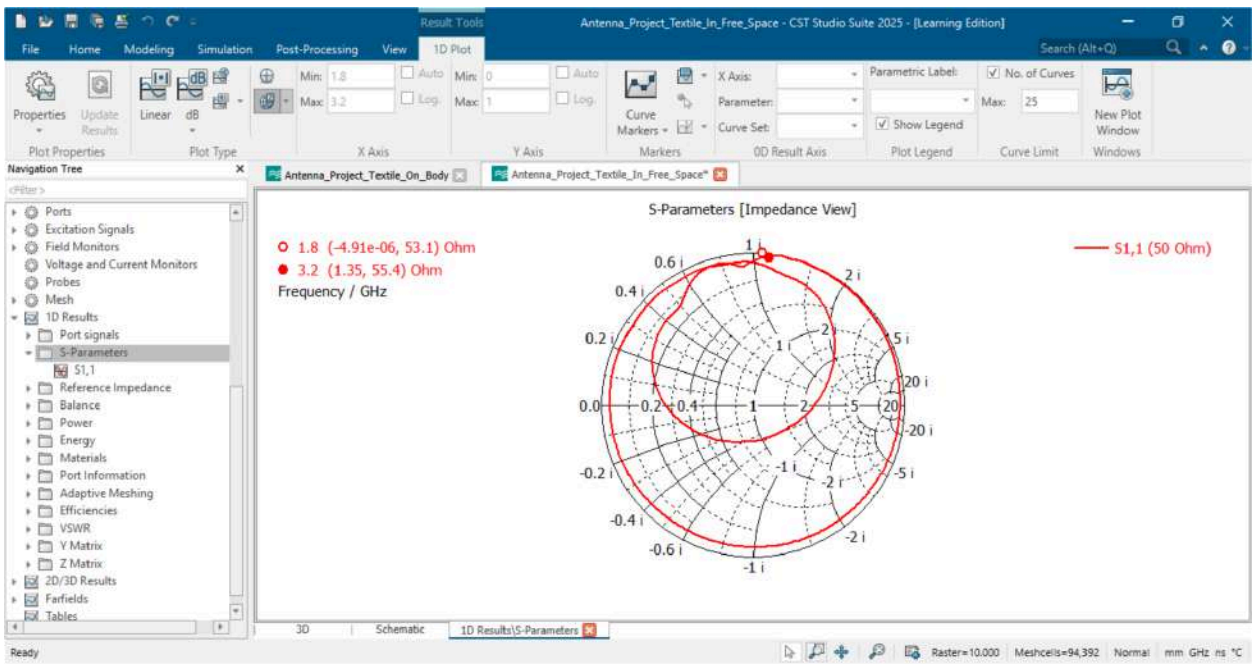


Figure 10 : Smith Chart Plot of the S Parameters for the Patch Antenna in Free-Space.

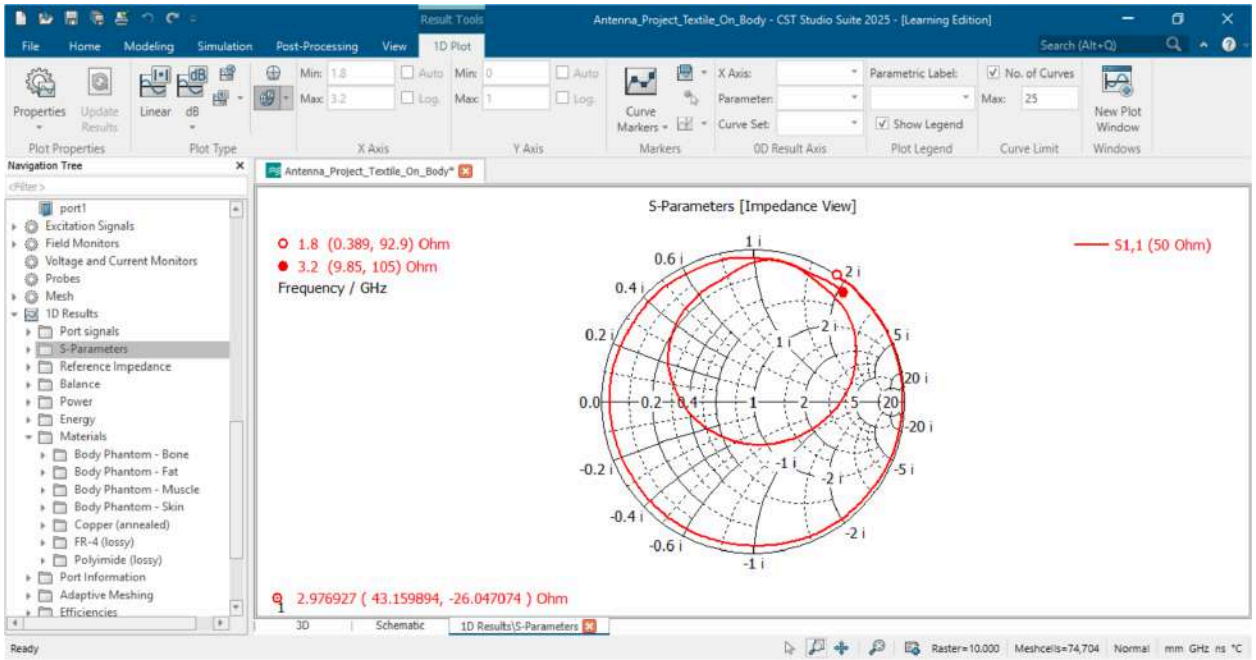


Figure 11 : Smith Chart Plot of the S Parameters for the Patch Antenna placed on the Body Phantom.

3. *VSWR Plot :*

   The Voltage Standing Wave Ratio (VSWR) graph gives us a measure of how effectively RF Power is transmitted from the feed line into the antenna. A VSWR that is less than 2 indicates acceptable performance. In free space the VSWR is around 1.6 at resonance which means that it demonstrates excellent matching. While, on body the VSWR ranges around 1.8 to 1.9 which is still within acceptable wearable antenna limits. Thus these results indicate that the antenna efficiently transfers power from the feed line to air without significant reflection.

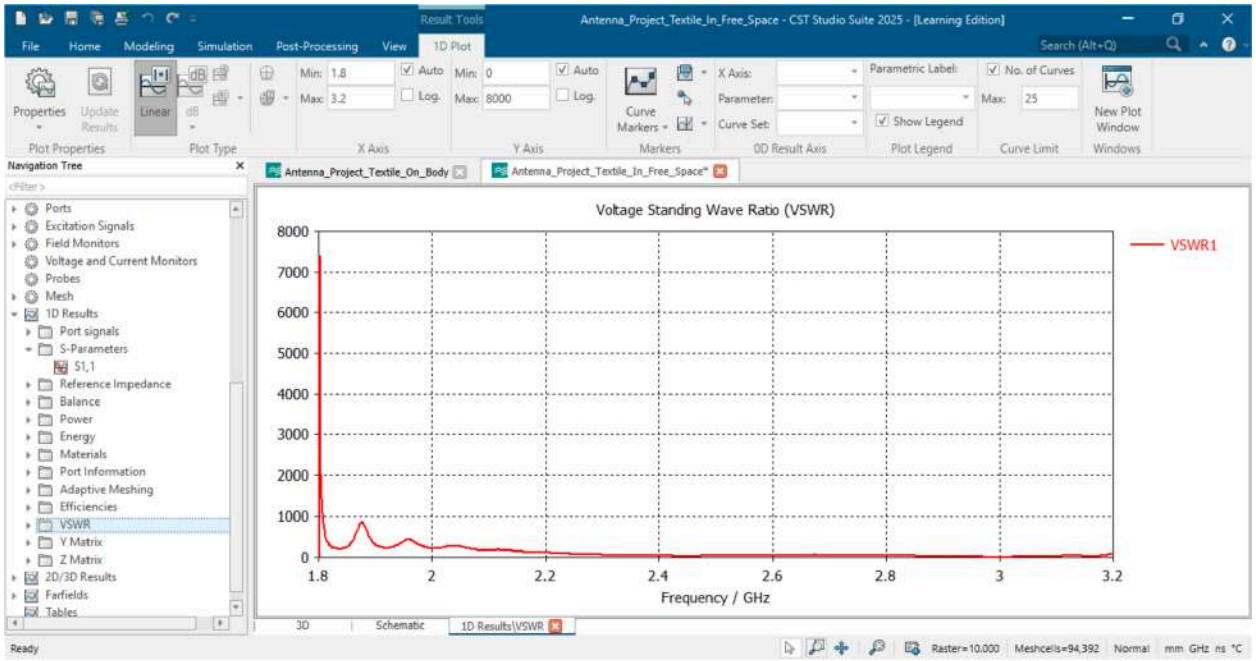


Figure 12 : VSWR Plot for the Patch Antenna in Free-Space.

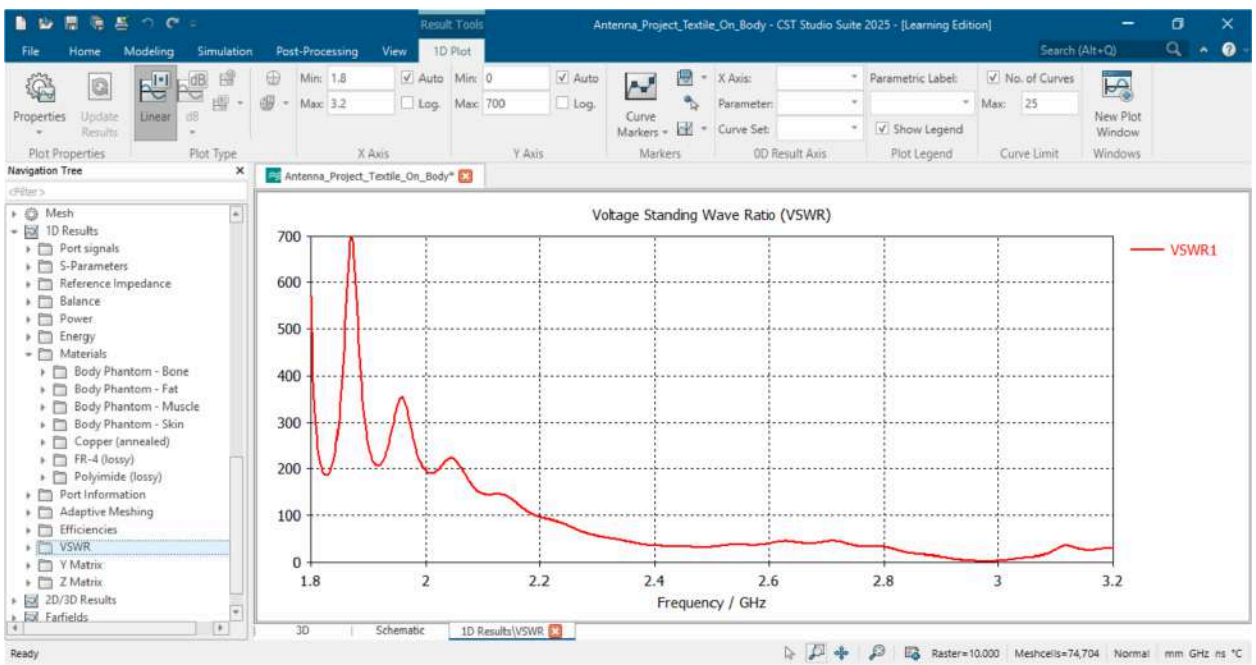


Figure 13 : VSWR Plot for the Patch Antenna placed on the Body Phantom.

4. *Far Field Directivity Abs Plot :*

   This plot helps us visualise how the radiated energy is distributed spatially. It quantifies the directivity, which indicates how concentrated the radiation is in a

particular direction. High directivity ensures effective signal transmission towards the desired region. The maximum directivity occurs along the normal axis of the patch surface. On body, there is slight distortion in the lobe pattern due to the antenna's interaction with the tissue which shows that some radiation is either absorbed or redirected but the overall shape remains almost the same for both.

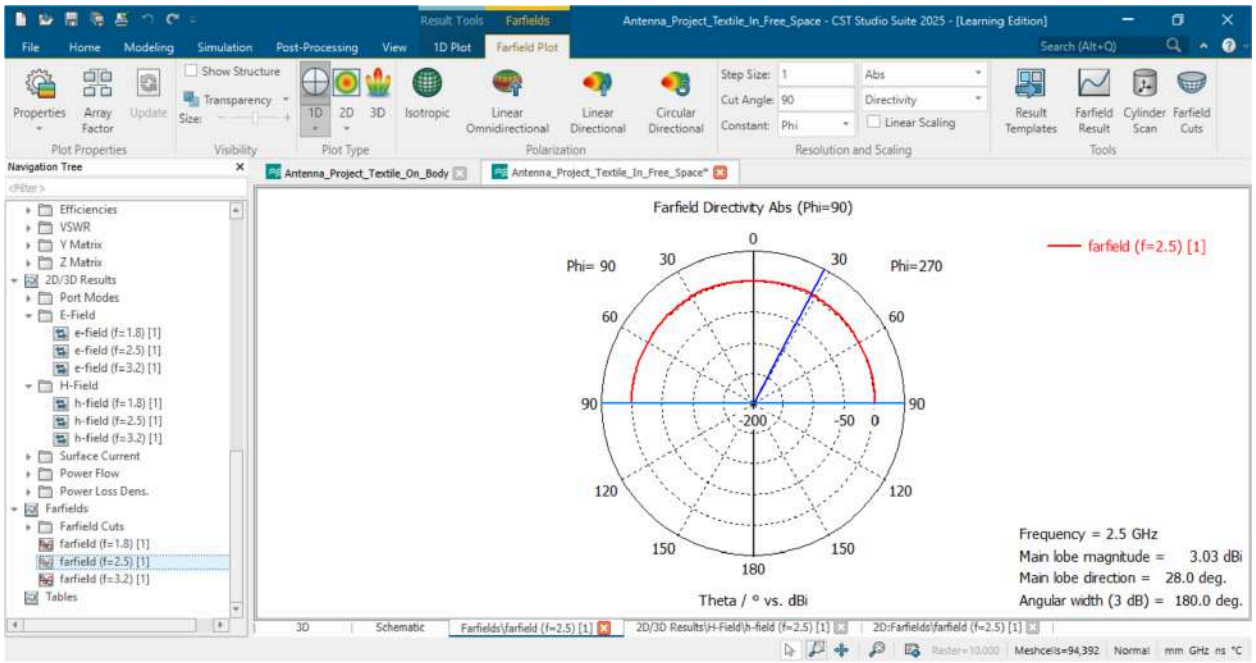


Figure 14 : Far Field Directivity Plot for the Patch Antenna in Free-Space.

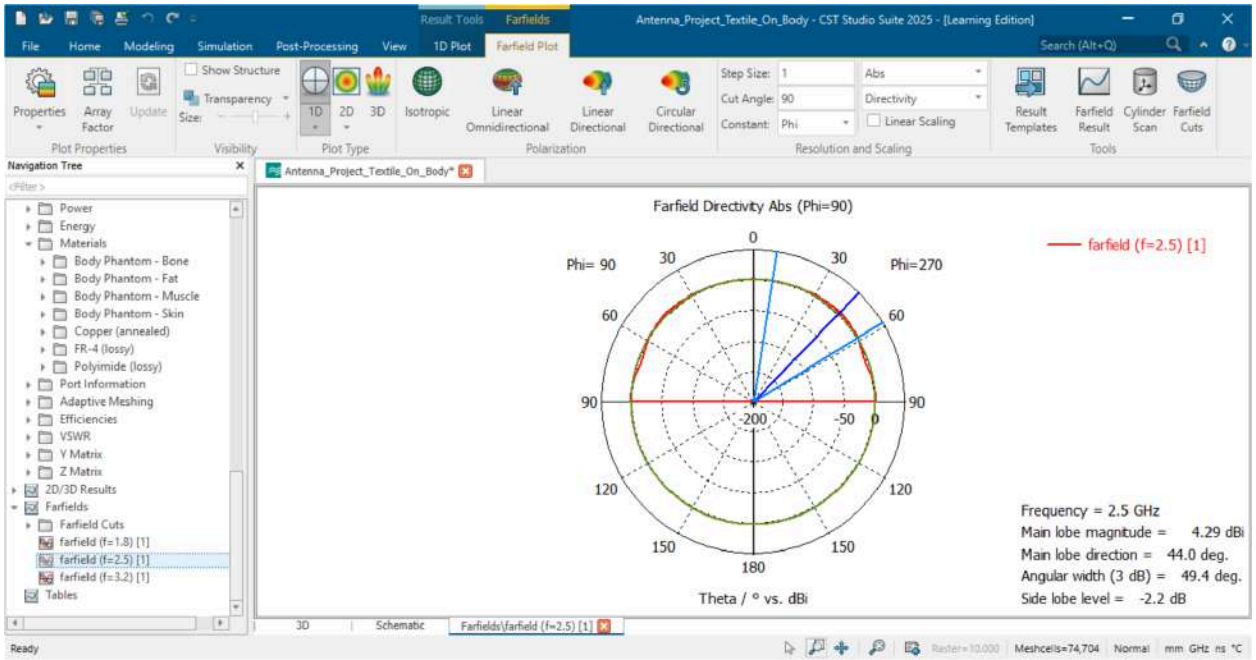


Figure 15 : Far Field Directivity Plot for the Patch Antenna placed on the Body Phantom.

5. *Efficiencies :*

   This plot reports the total radiation efficiency, accounting for conductor, dielectric, and mismatch losses. It is crucial for wearable antennas where the proximity to the lossy human tissue can significantly reduce efficiency in comparison to the efficiency we obtained from the simulation carried out in free space.

   In our study, the antenna achieves high efficiency in free space. When placed on the body, the efficiency drops slightly because a little part of the radiated power is absorbed by the tissues. This shows that the dielectric loading affects performance but does not critically degrade it.

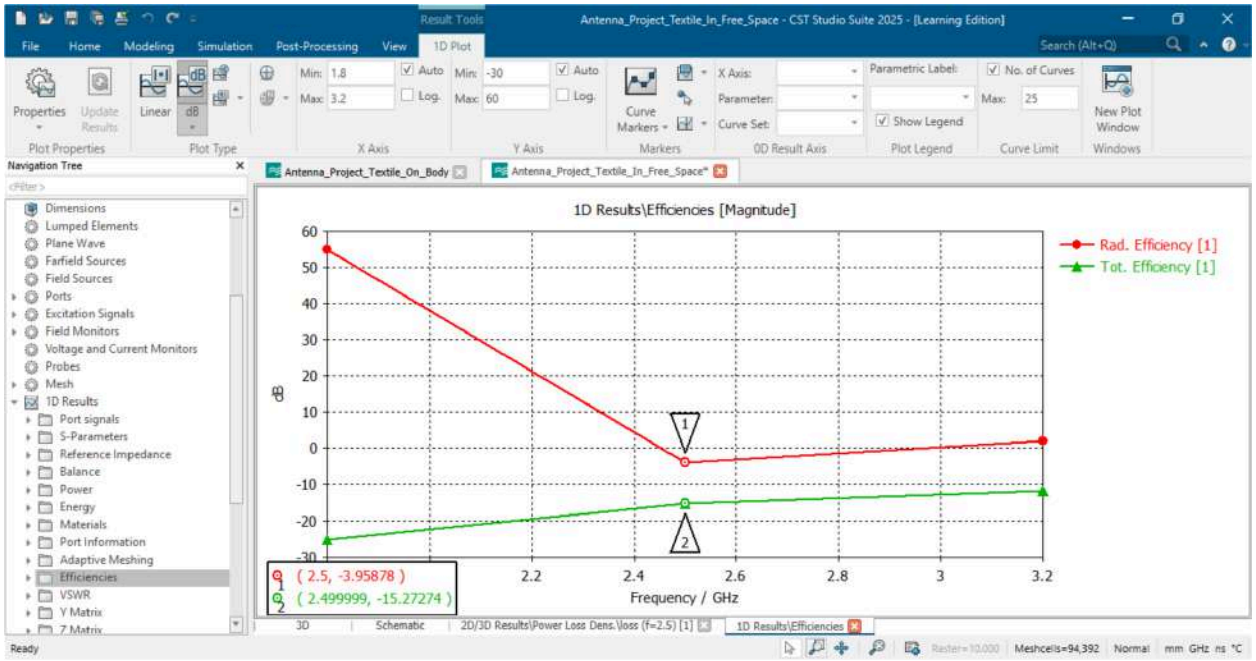


Figure 16 : Efficiencies Plot for the Patch Antenna in Free-Space.

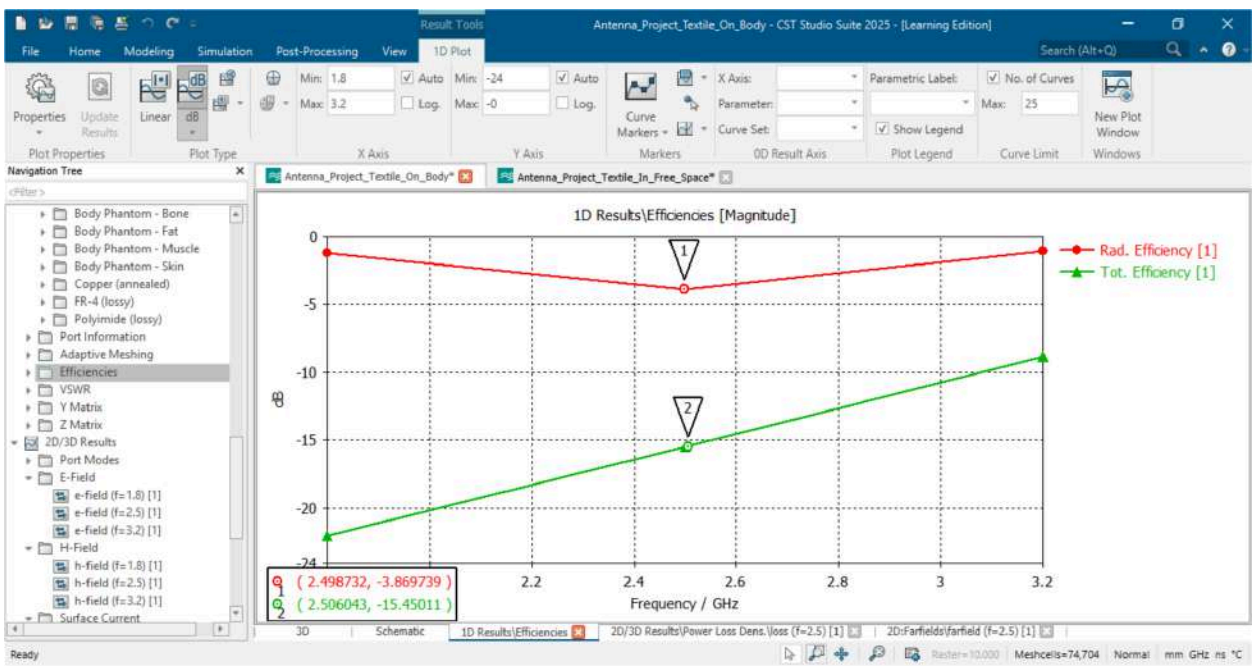


Figure 17 : Efficiencies Plot for the Patch Antenna placed on the Body Phantom.

6. *E-Plane 2D Distribution :*

   The E-Plane 2D radiation pattern shows how the antenna radiates in the plan containing the electric field and the direction of maximum radiation. It helps identify the bean shape, load formation and the side-lobe levels which are important for

assessing directional stability.

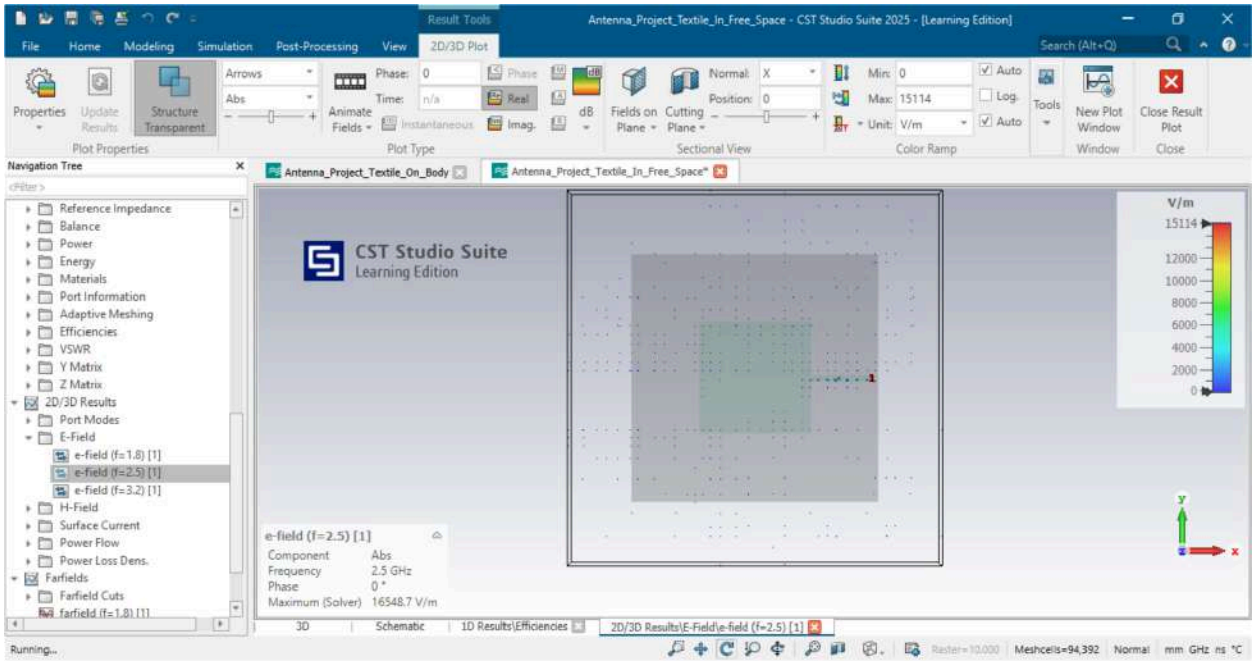


Figure 18 : 2D E-Plane Distribution for the Patch Antenna in Free-Space.

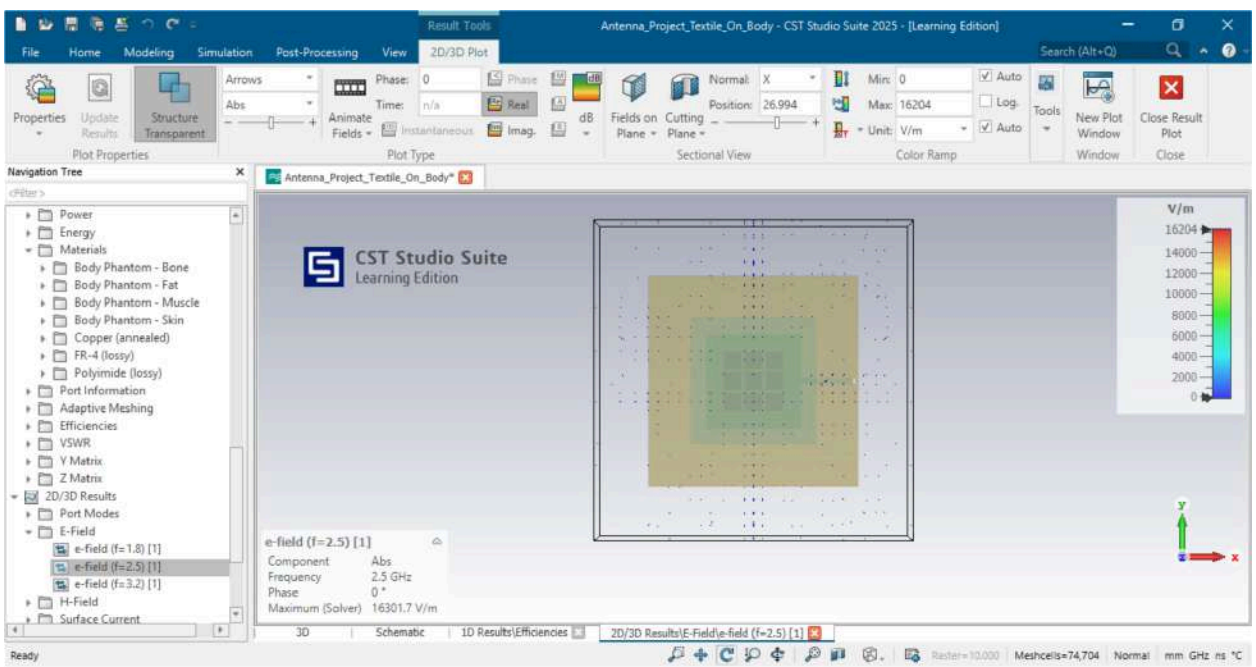


Figure 19 : 2D E-Plane Distribution for the Patch Antenna placed on the Body Phantom.

7. *E-Fields through the cut :*

The field-cut visualization displays the electric-field intensity within the antenna

structure and near its surface. It gives us an insight into the field confinement, energy leakage, and interaction with nearby dielectric materials such as the human body.

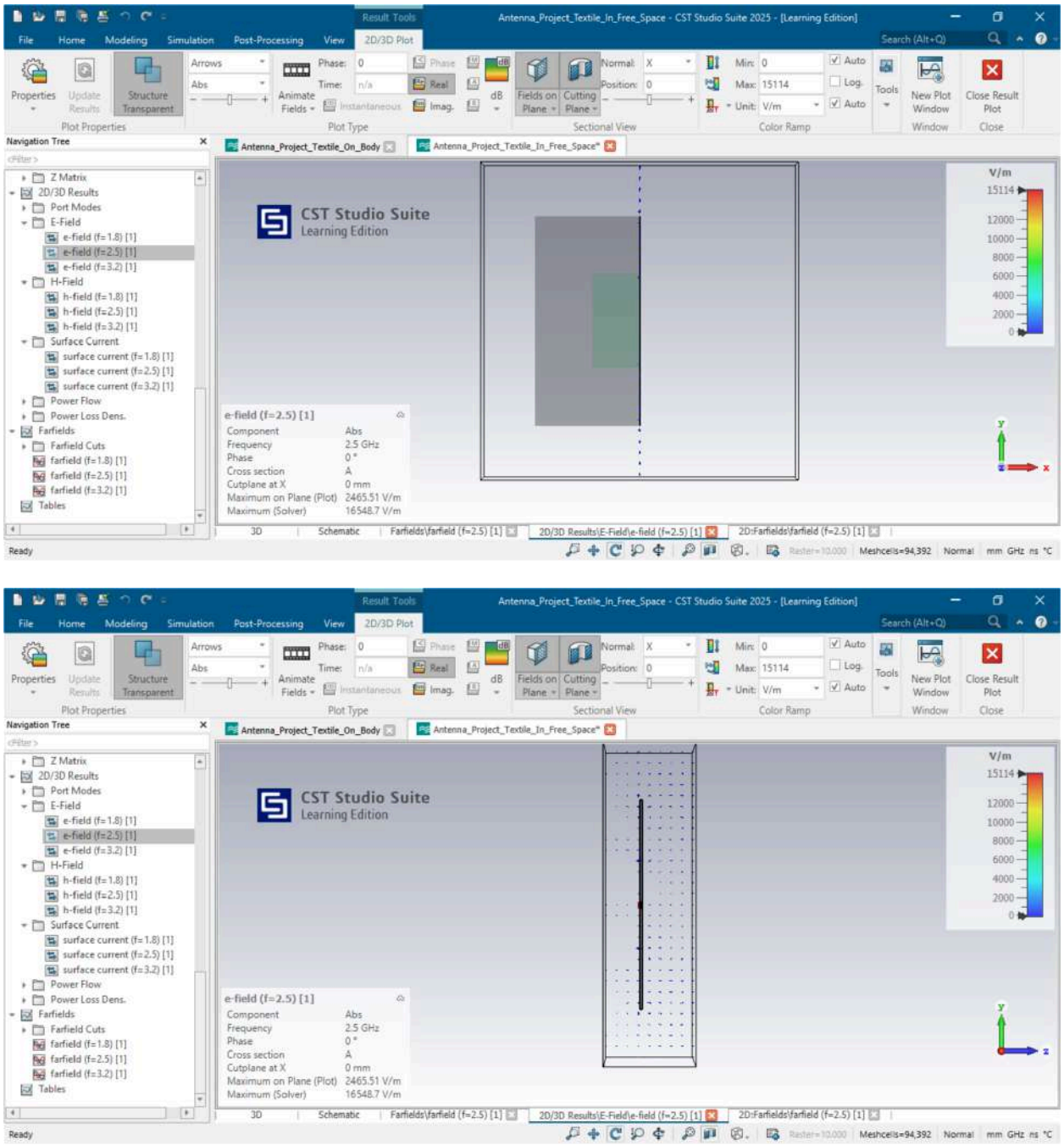


Figure 20 : 2D E-Plane Distribution through the cut for the Patch Antenna in Free-Space.

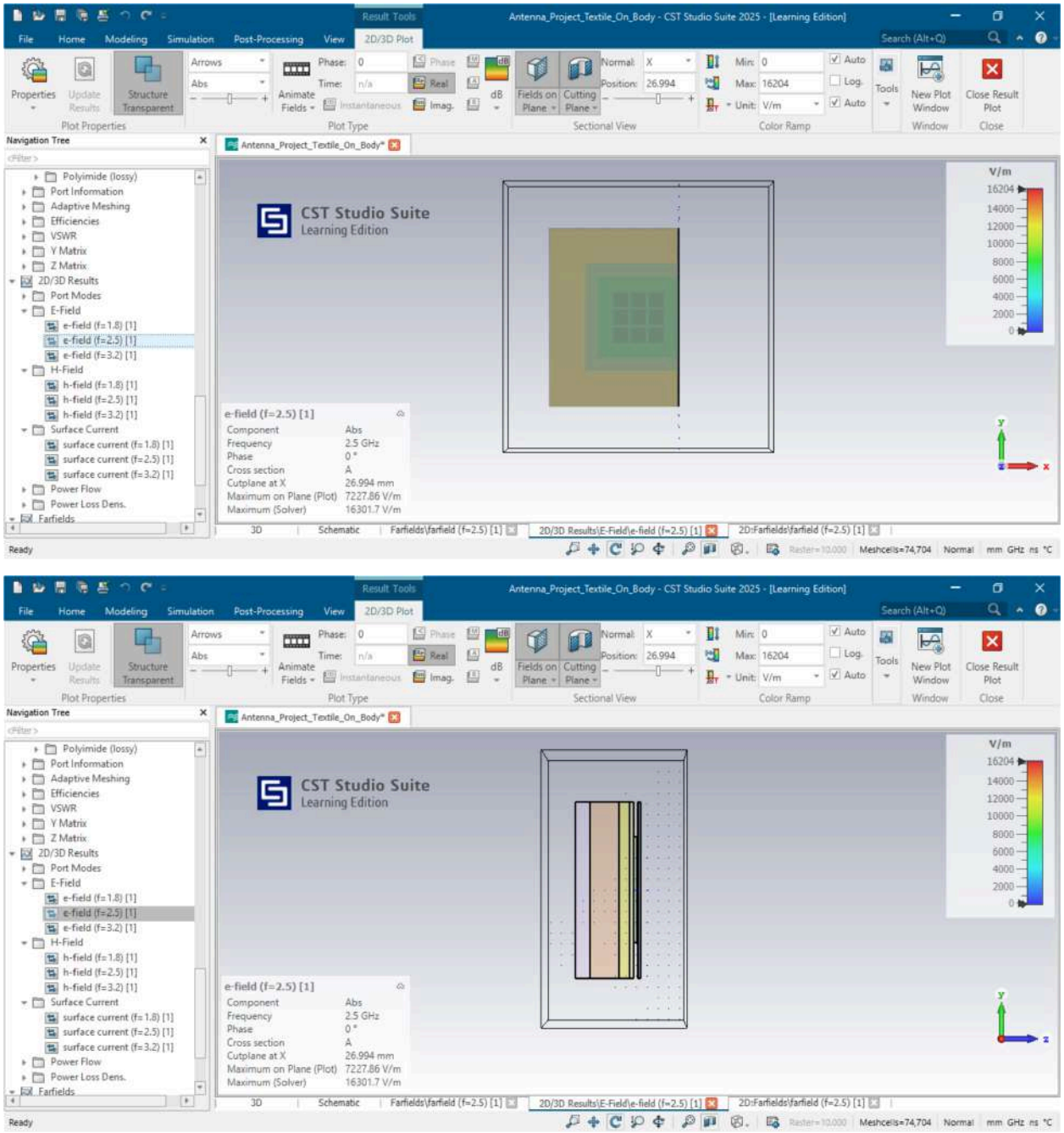


Figure 21 : 2D E-Plane Distribution through the cut for the Patch Antenna placed on the Body Phantom.

8. *H-Plane 2D Distribution :*

   The H-Plane pattern shows the magnetic-field distribution in the plan perpendicular to the E-Plane. Together with the E-Plane plot, it helps us ensure that the antenna radiates symmetrically and verifies polar characteristics

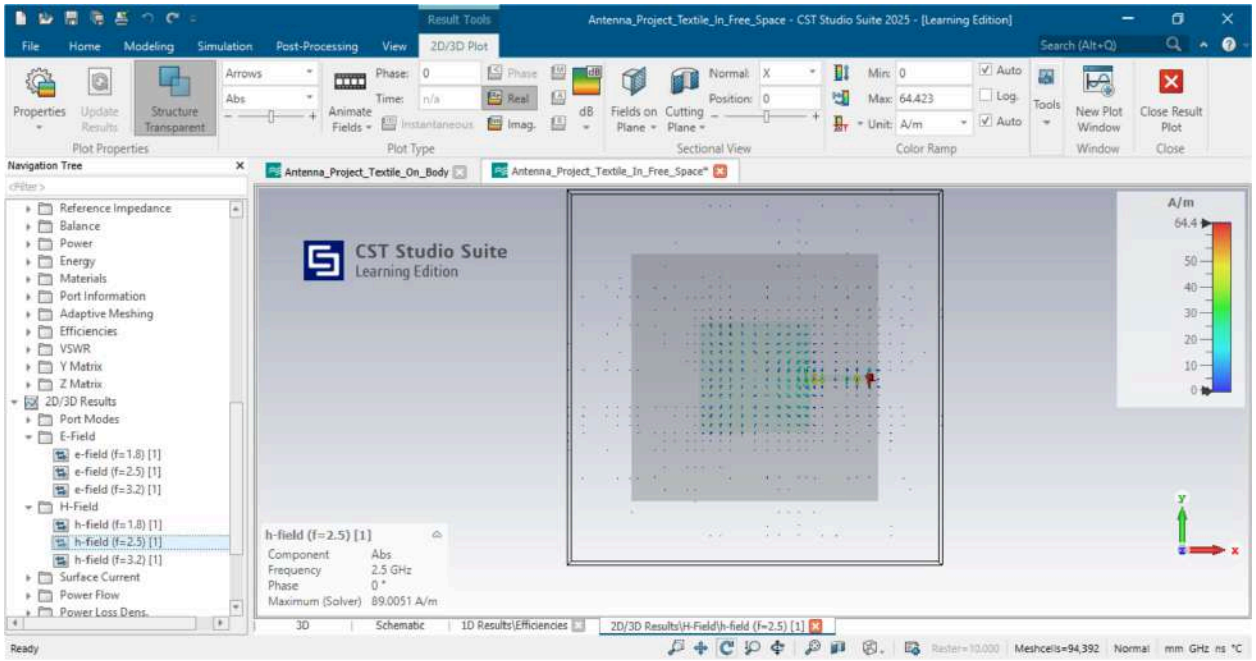


Figure 22 : 2D H-Plane Distribution for the Patch Antenna in Free-Space.

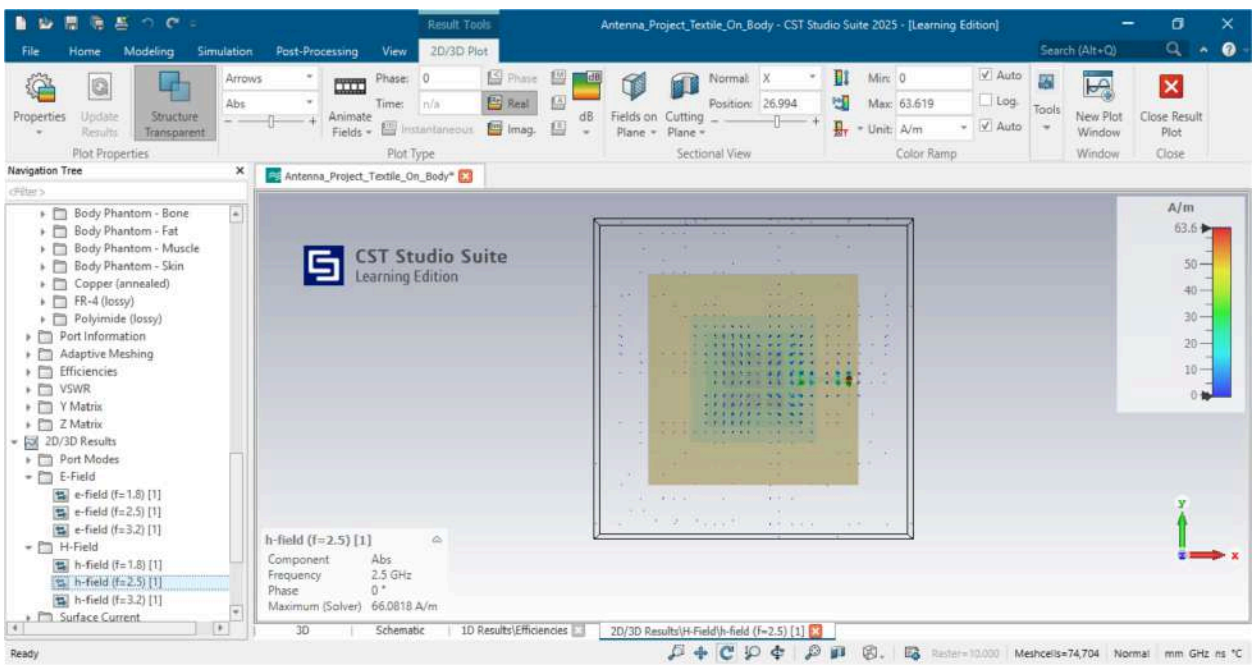


Figure 23 : 2D H-Plane Distribution for the Patch Antenna placed on the Body Phantom.

9. *H-Fields through the cut :*

The field-cut visualisation reveals the magnetic-field strength distribution across the structure which helps visualize the complementary behavior of the magnetic fields

relative to the electric fields. It confirms proper excitation of the intended resonant mode and highlights potential coupling to the body tissue.

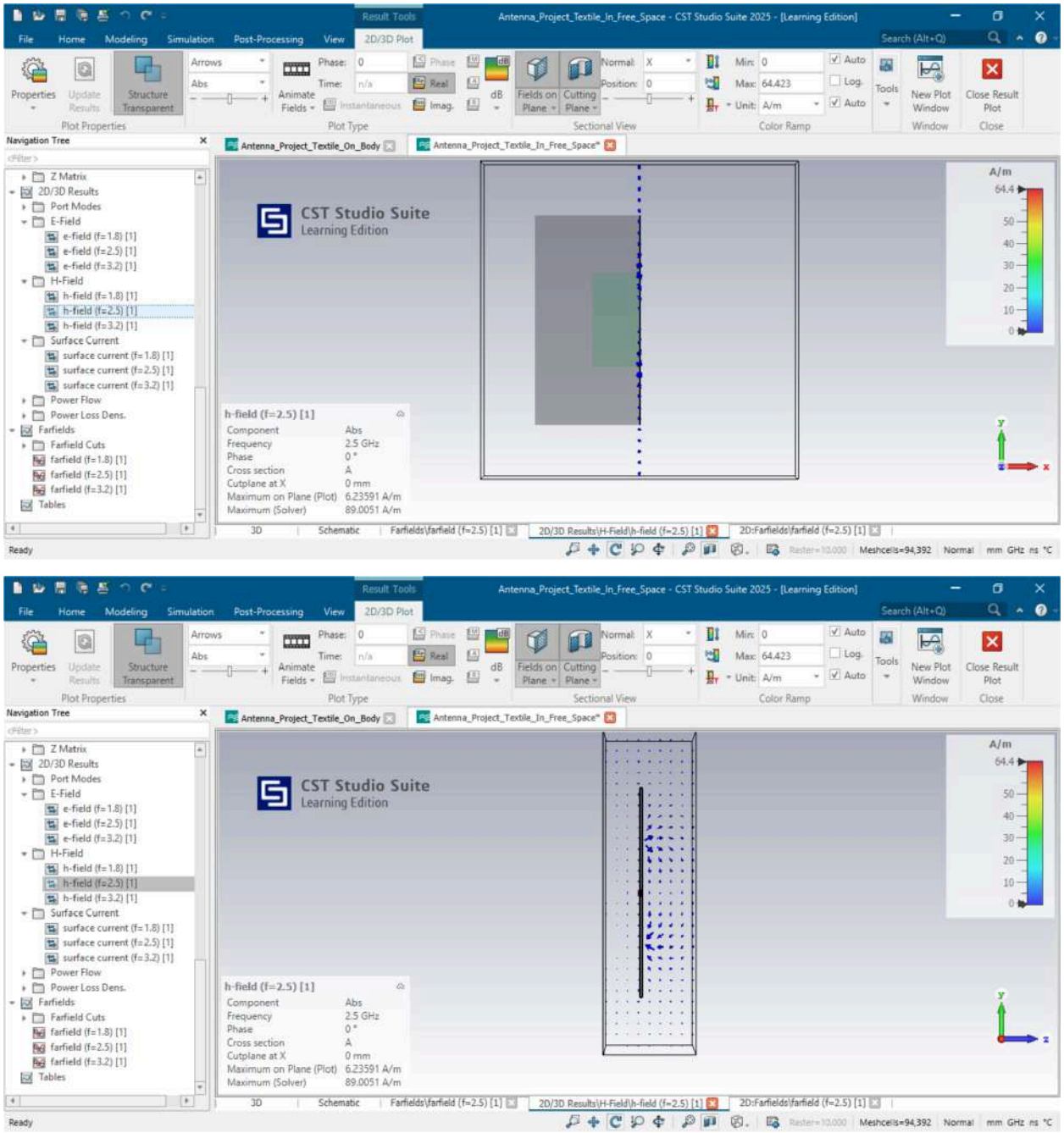


Figure 24 : 2D H-Plane Distribution through the cut for the Patch Antenna in Free-Space.

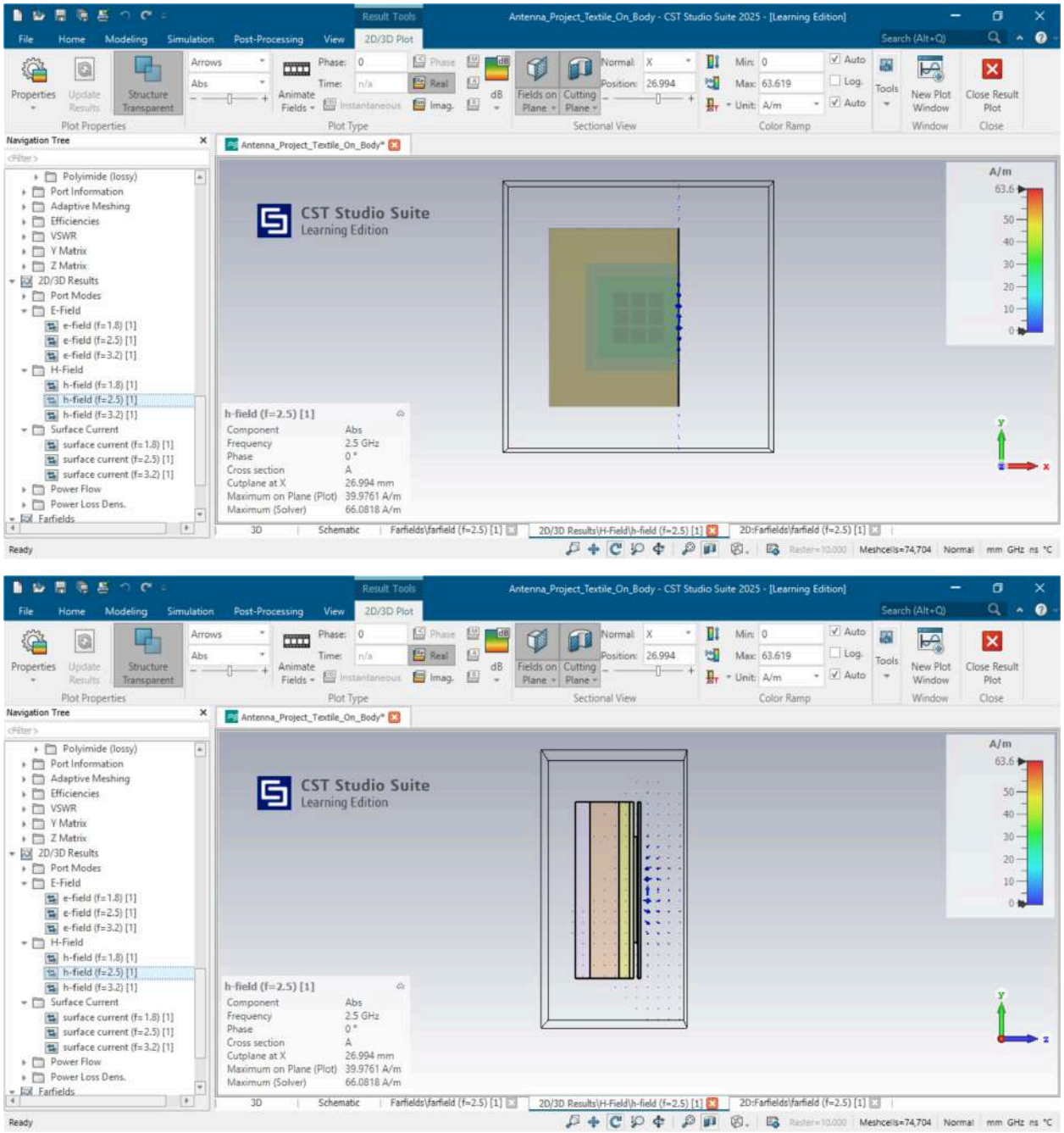


Figure 25 : 2D H-Plane Distribution through the cut for the Patch Antenna placed on the Body Phantom.

The plots that we have obtained for the E-Plane and H-plane distributions confirm that the antenna radiates primarily in the forward direction with a peak gain of around 3dBi in free space. On the body, the radiation becomes slightly less directional but remains functional, which is an ideal characteristic for wearable devices, particularly ones that do not require high focused beams.

We can also see that the electric field is the strongest at the edges of the patch, aligning with the current null regions. On the body, the field lines extend slightly into the body model which indicates an SAR rise, but also tells us that it is within safe limits. It helps us visually confirm how the antenna interacts electromagnetically with the human tissue.

The magnetic fields seem to form closed loops below the patch and spread into the surrounding space. The on-body simulation shows a slight field compression near the phantom surface which is a typical performance description for wearable antennas operating near lossy materials.

10. *2D Far Field Distribution :*

The 2D far-field radiation plot maps the radiation pattern on a single plane at the resonant frequency. It helps in comparing the beamwidth, null directions, and side lobes in case of both the conditions.

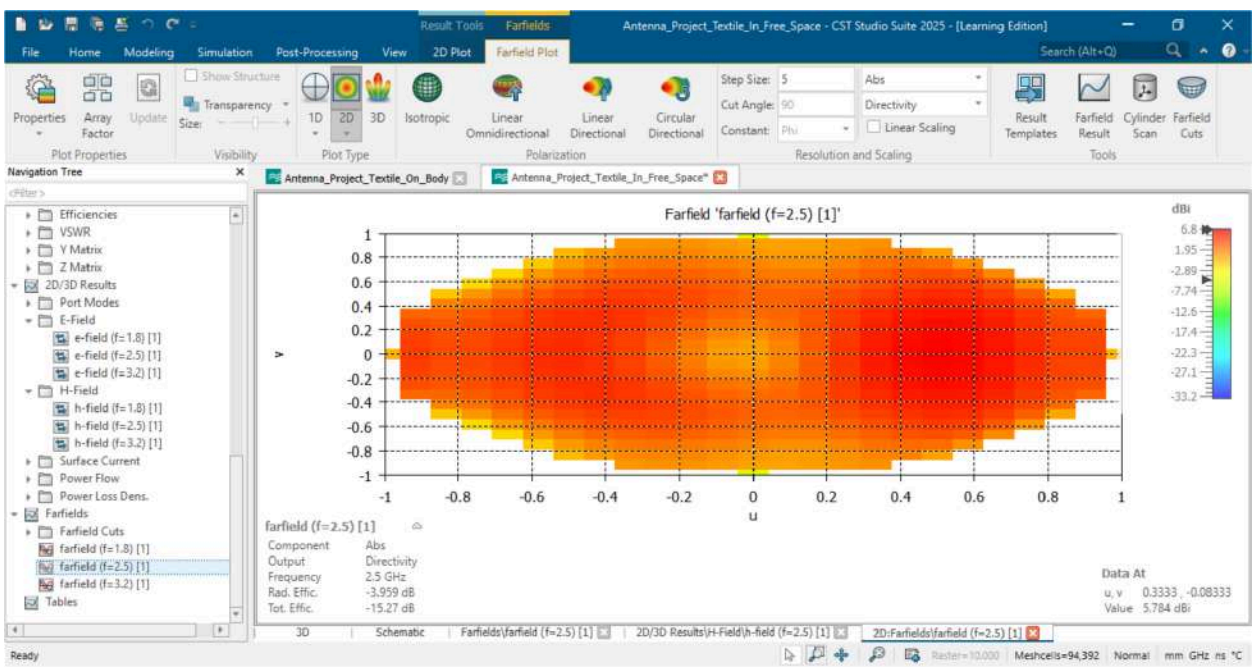


Figure 26 : 2D Far Field Distribution for the Patch Antenna in Free-Space.

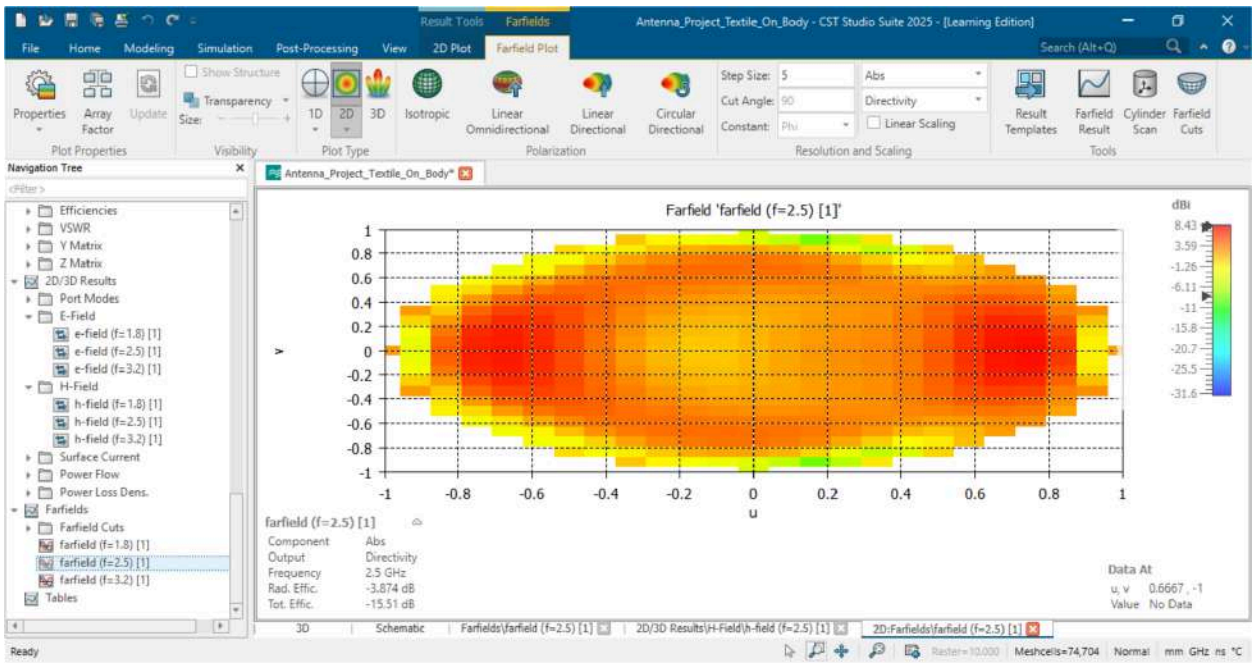


Figure 27 : 2D Far Field Distribution for the Patch Antenna placed on the Body Phantom.

11. *3D Far Field Distribution :*

The 3D radiation pattern provides a comprehensive view of the overall radiation characteristics of the antenna. It visualizes the main lobe direction, gain distribution, and how body loading affects beam symmetry or causes backward radiation.

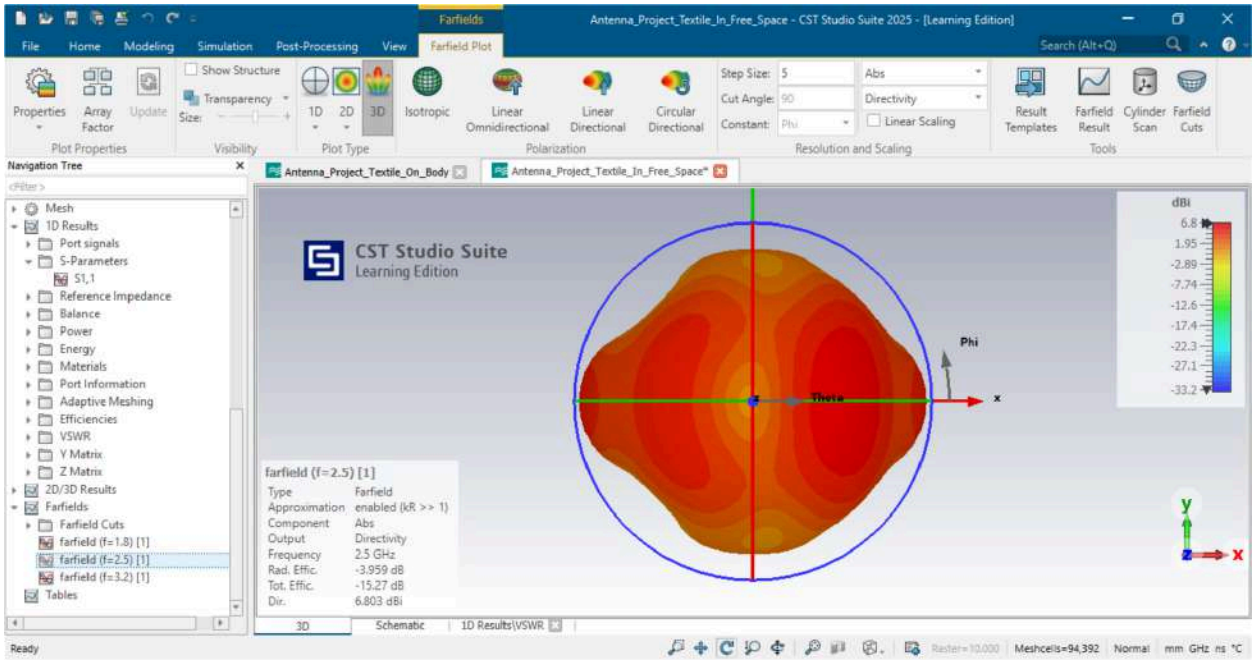


Figure 28 : 3D Far Field Distribution for the Patch Antenna in Free-Space.

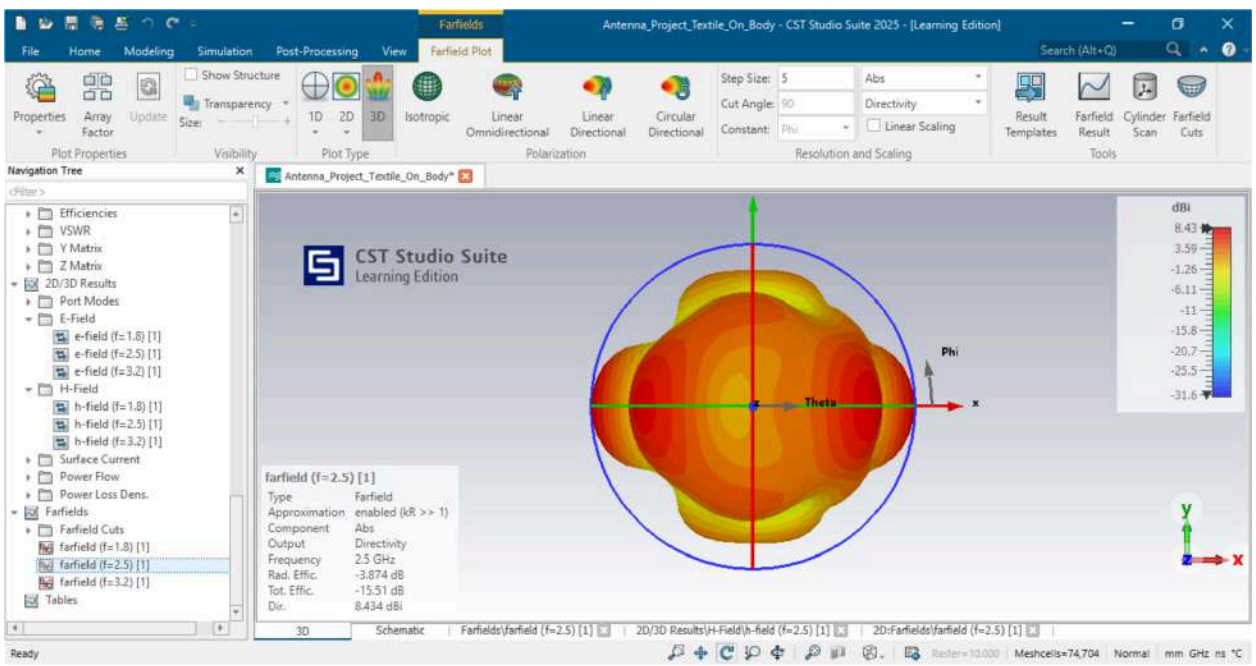


Figure 29 : 3D Far Field Distribution for the Patch Antenna placed on the Body Phantom.

12. *Surface Current Map :*

The surface current map shows the current density distribution on the patch and the ground place. It identifies the active regions contributing to radiation and helps

validate the resonant mode. Proper excitation is indicated by an increased current at the feed and edges. From the simulations, we can observe that the current is concentrated along the edges of the path, especially near the feed and the radiating slots which confirms the correct excitation mode. When placed on the body, the current magnitude slightly reduces due to the lossy loading from the body phantom, but the path of the current distribution remains consistent which indicates stable operation.

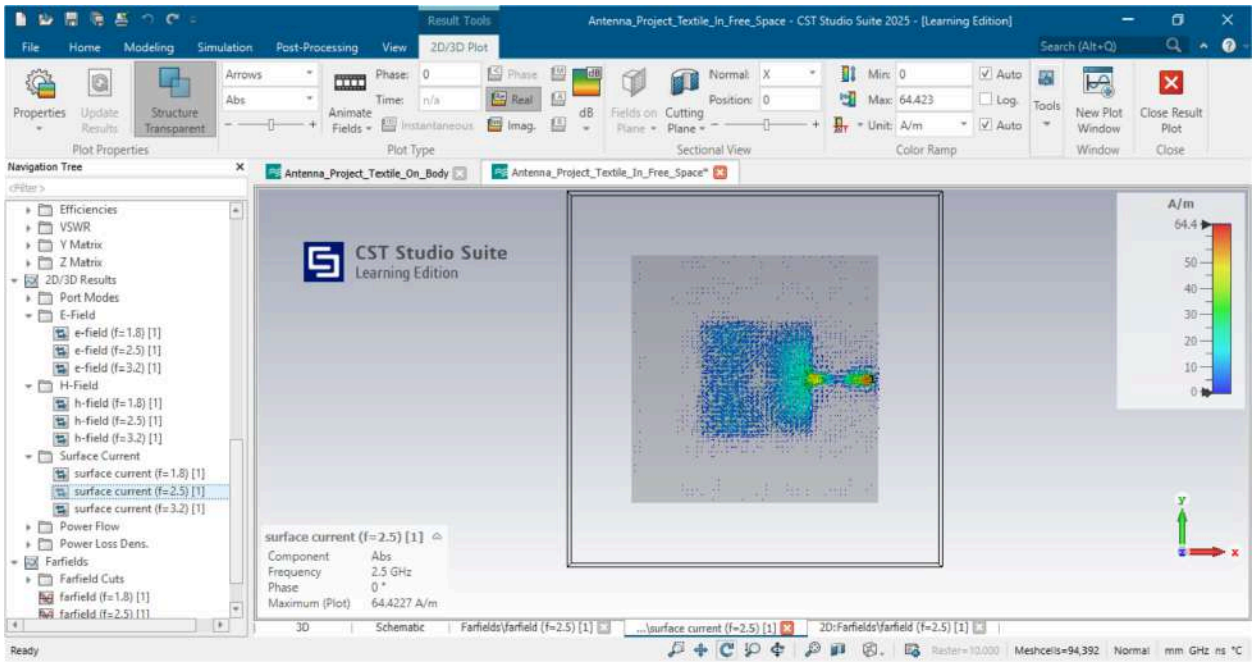


Figure 30 : Surface Current Distribution for the Patch Antenna in Free-Space.

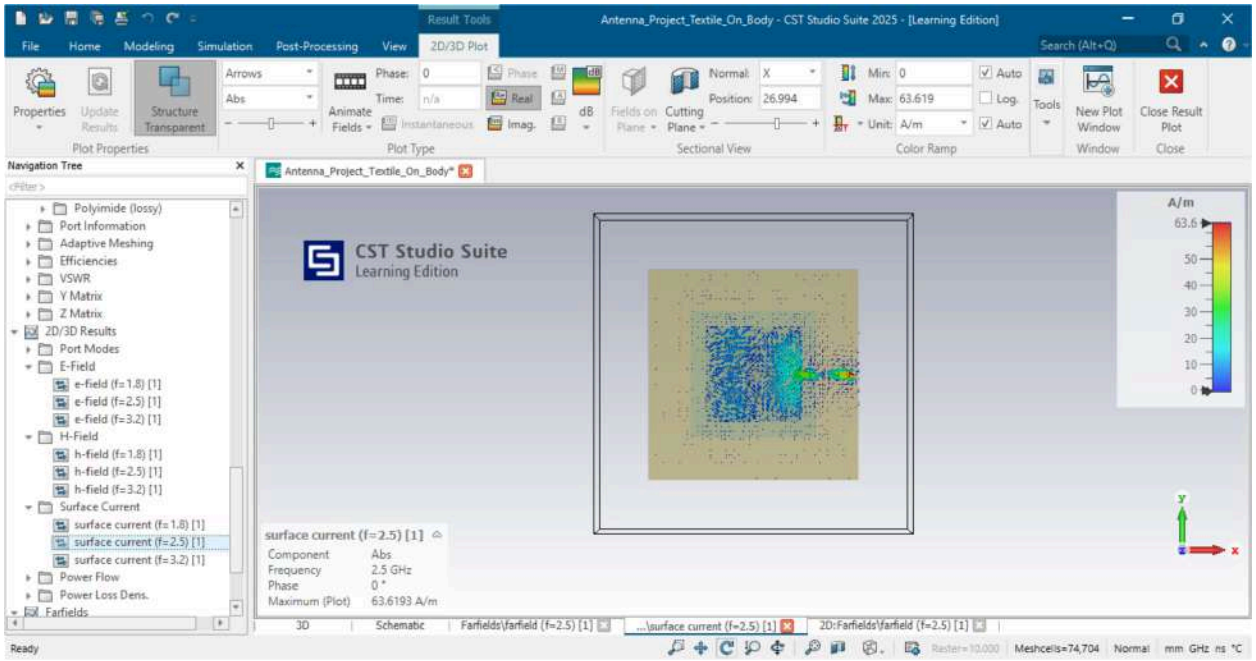


Figure 31 : Surface Current Distribution for the Patch Antenna placed on the Body Phantom.

13. *Power Loss/SAR Analysis of the Patch Antenna at 2.5GHz :*

SAR or Specific Absorption Rate distribution visualises the energy absorption within the body phantom at the operating frequency. It is essentially studied to evaluate the user's safety and verifying its compliance with exposure standards. The peak absorption occurs directly under the antenna center, which is an expected result. This proves that the antenna is safe for human exposure and suitable for wearable biomedical applications. High-intensity regions near the skin indicate where electromagnetic power converts to heat, guiding design improvements such as AMC backing or spacing. Our design layers consist of an AMC backing layer which helps us enhance the power loss and energy absorption to the skin.

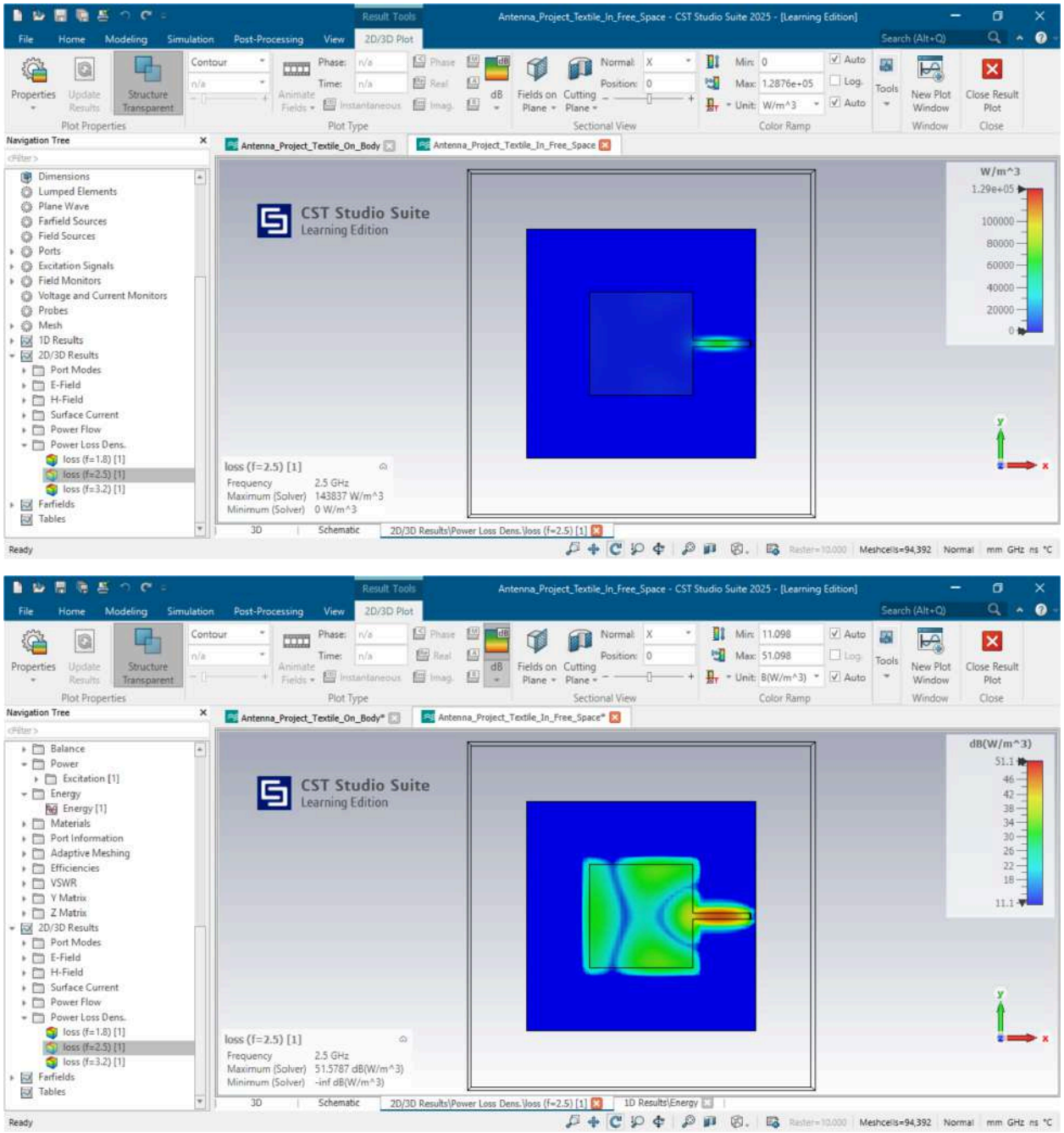


Figure 32 : Power Loss/SAR Distribution for the Patch Antenna in Free-Space.

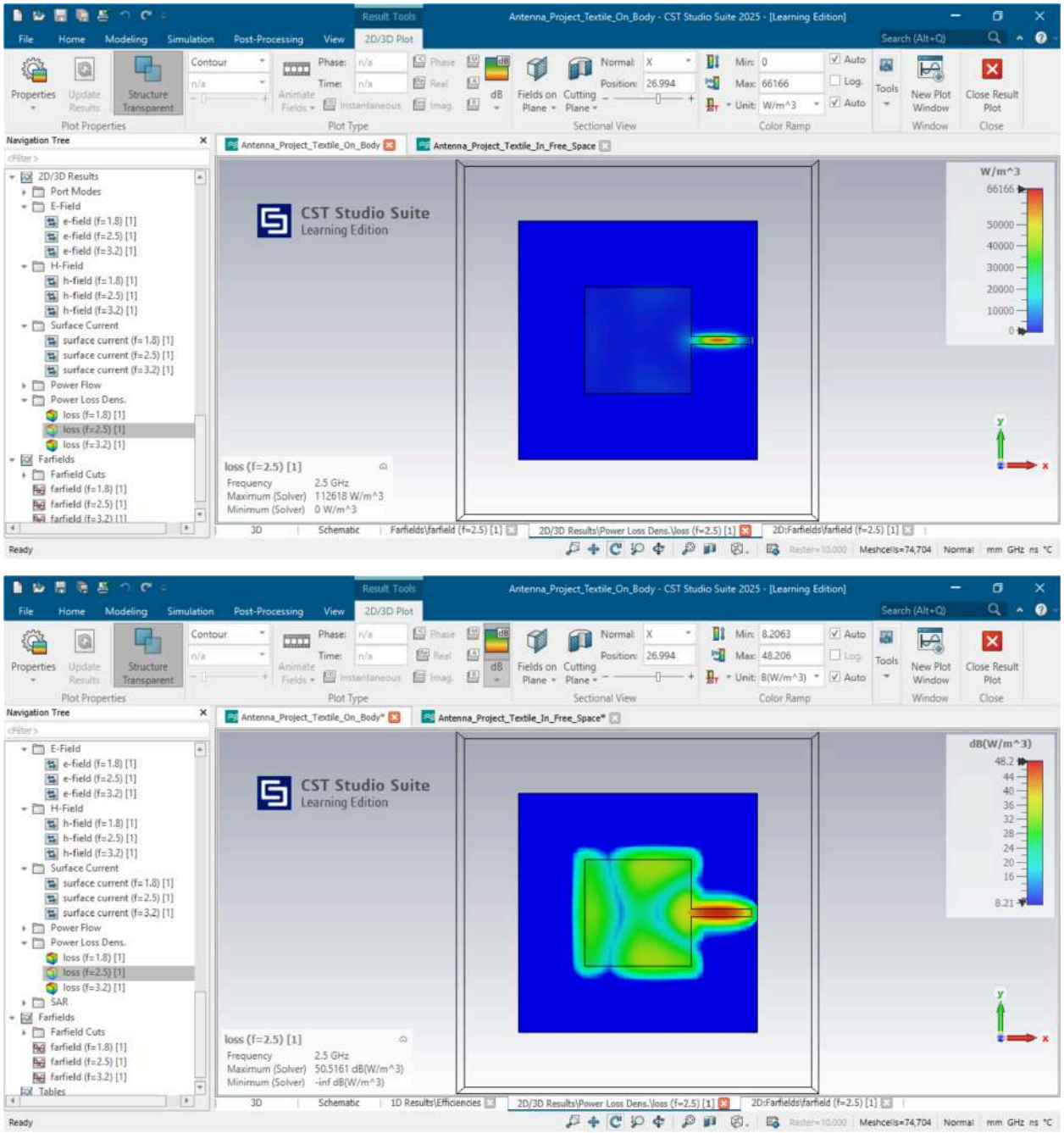


Figure 33 : Power Loss/SAR Distribution for the Patch Antenna placed on the Body Phantom.

The comparative analysis of the radiation pattern of the antenna in both environments shows that the proposed antenna maintains stable and efficient operation under both conditions. A minor resonant frequency shift and slight reduction in return loss and efficiency are observed due to the body-tissue loading, but the overall performance remains within the wearable limits. The radiation and current distributions confirm consistent excitation and directional stability, while the SAR analysis validates the user's safety. These results demonstrate that the

antenna achieves reliable impedance matching, effective radiation, and safe power absorption, proving its suitability for biomedical and wearable communication applications.

6. CONCLUSION AND FUTURE WORK

6.1 SUMMARY :

The proposed project successfully demonstrates the design and simulation of a flexible microstrip patch antenna for wearable biomedical applications, operating at the 2.45 GHz ISM band. The antenna was modeled on a felt substrate ($\varepsilon r \approx 1.8$, loss tangent = 0.02), and analyzed in both free-space and on-body configurations using CST Studio Suite. In free space, the antenna achieved resonance near 2.99 GHz with $S_{11} = -12.77$ dB, while on-body testing showed a minor shift to 2.97 GHz with $S_{11} = -11.09$ dB, maintaining good impedance matching (VSWR ≈ 1.8). On a comparative analysis the radiation patterns remained stable, and SAR evaluation at 2.5 GHz confirmed values within safe exposure limits, validating the user's safety. These results confirm that the designed antenna provides efficient radiation, mechanical flexibility, and biocompatibility, making it a strong candidate for integration into wearable health-monitoring systems and future WBAN applications.

6.2 LIMITATIONS AND CONSTRAINTS :

Although the study yielded positive results, a number of limitations were also noted while performing this comparative analysis :

- The design and validation were limited to CST Studio Suite simulation environments; neither physical fabrication nor experimental testing were conducted which are required to study the antenna's performance in real-world environments.

- The simulation did not account for environmental factors like temperature, humidity, and perspiration, which are common in real-world wearable scenarios.

- The SAR estimation was based on a simplified tissue phantom, which might not accurately represent multilayer human tissue diversity.

6.3 IMPROVEMENT/FUTURE WORK :

The project can be further worked upon and enhanced by :

- *Physical prototyping and measurement :* Fabricating the antenna using conductive textiles and performing experimental validation of gain, return loss, and SAR to compare with simulated data.

- *Material innovation :* Exploring graphene-based or conductive polymer textiles that will combine high conductivity with improved mechanical flexibility.

- *Environmental robustness testing :* Assessing antenna performance under varying humidity, sweat exposure, and temperature to ensure real-world reliability.

- *Advanced modeling :* Implementing multilayer, anatomically accurate human phantoms to help better estimate SAR and detuning effects.

- *Integration with sensors :* Coupling the antenna with real-time biosensors (ECG, SpO_2, temperature) to form a complete wearable health-monitoring system.

- *Miniaturization and optimization :* Using metamaterials, AMC/EBG surfaces, or SIW-based designs to further enhance the isolation and efficiency while reducing size.

7. SOCIAL AND ENVIRONMENTAL IMPACT

The development of a flexible, textile-based wearable antenna directly contributes to the advancement of personalized healthcare by enabling continuous, non-invasive health monitoring through wireless body area networks.

Socially, this technology can enhance accessibility to healthcare, particularly for the elderly, chronically ill, or individuals in remote areas, by transmitting vital physiological data in real time to healthcare professionals. It promotes preventive medicine, early diagnosis, and patient-centered care.

From an environmental perspective, the use of textile substrates such as felt and conductive fabrics aligns with the principles of sustainable electronics and SDG 12 (Responsible Consumption and Production). Textile-based antennas can be fabricated using eco-friendly materials and manufacturing techniques with minimal electronic waste. Moreover, flexible antennas consume less raw metal and can be integrated into garments, reducing plastic-based packaging and housing materials. With proper end-of-life recycling strategies, they can significantly lower the environmental footprint of wearable electronics.

8. WORK PLAN

8.1 : TIMELINE :

	September	October	November
Finalising the Project Title			
Conducting the Literature Review			
Initial Findings and Finalizing on the Design Approach			
Final Simulation on the CST Studio Suite Software			

8.2 INDIVIDUAL CONTRIBUTION :

Individually each one of the team members has made good contributions in order to accomplish the outcomes and objectives of the project.

1. The finalized project title was chosen and the project outcomes were decided by Mugdha, Varuni and Srijita after going through several review and conference papers.
2. The literature review was conducted together by Varuni and Mugdha.
3. The initial findings and finalization on the design approach was done by Yuval.
4. The final simulation and analysis of the antenna on the CST Studio Suite Software was done by Ishita.

9. COST ANALYSIS

Since the project was conducted entirely through CST Studio Suite simulations, no additional hardware or fabrication costs were minimized, making the design stage highly cost-efficient for preliminary validation. For further implementation and fabrication, the primary costs would arise from antenna fabrication, material procurement, and testing.

10. PROJECT OUTCOME

By conducting this project we have attained the following outcomes :

- Successfully designed and simulated a flexible textile antenna for 2.5 GHz ISM applications using CST Studio Suite.
- Verified that the antenna demonstrates a $S_{11} < -10$ dB, VSWR ≈ 2, and consistent gain across flat and bent configurations.
- Demonstrated a good SAR compliance (< 1.6 W/kg) ensuring safe human-body usage.
- Performed a comparative analysis between free-space and on-body performance of the antenna, highlighting minimal detuning and stable radiation patterns.
- Contributed towards future wearable biomedical systems, proving that flexible textile substrates can support efficient wireless communication.

Overall, the project establishes a strong foundation for fabric-integrated, safe, and efficient wireless antennas suited for next-generation wearable health-monitoring systems.

11. REFERENCES

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